Which Side Is Wrong? What Contradiction Flags Do to Retrieval-Augmented Generation Under Misinformation

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Abstract

Retrieval-augmented generation systems increasingly detect contradictions among retrieved passages and flag them to the generator. We ask what such a flag must say, and how accurate its detector must be, for it to help against misinformation. On fixed passages from five benchmarks, with seven generators from three model families, we vary the detector and the format of its flag, in hypotheses pre-registered in stages (H1–H9 before any confirmatory data, H10–H31 in eight later addenda). A perfectly accurate neutral flag ("these passages conflict") does not reliably help: what it does depends on the answer policy and the model family, and pooled over 34 settings its effect is -1.6 points $[-5.5, +2.2]$. Naming the false side helps, but no more than stating the wrong answer outright (pooled difference -0.1 points, equivalent within ± 3). In a randomised test, a flag naming the false side raises accuracy by $+17.6$ to $+33.5$ points and one naming the correct side lowers it by 31.2 to 64.5, so a directional flag helps only when the detector names the false side more often than a break-even rate that rises with the generator's accuracy, from 0.52 to 0.79. The follow rates measured in this test predict the flag's effect in 30 crossed detector–generator cells ($r = 0.94$, mean error 1.8 points; exploratory), including where an imprecise detector hurts strong generators. What a detector contributes is its verdict: passed as an answer key it matches the flag within 3 points. On genuine conflicts direction removes valid answers, and when conflicts are rare a real detector's false flags on clean questions can outweigh its gains. We release the harness, the pre-registrations and every run.

1 Introduction

Retrieval-augmented generation (RAG) (Lewis et al., 2020) hands a language model passages that may disagree. Some disagreements are genuine; others are planted or stale misinformation (Zou et al., 2025; Wang et al., 2025b). A natural fix is to detect the contradiction and tell the generator. Recent systems pair contradiction detectors with prompts that flag conflicting passages (Wang et al., 2026), and prompting a model about conflicts can help (Cattan et al., 2025). These systems report detection F1 and end-to-end accuracy, but not how the two connect. What should the flag say, and what must the detector get right for it to help?

We fix the retrieved passages and vary, independently, the *detector* (heuristic, NLI, LLM, random placebo, oracle, and synthetic detectors of chosen quality) and the *format* that passes its output to the generator: a neutral "these passages may conflict" flag, a flag that says one side is false without saying which, a *directional* flag that names the likely misinformation, or removal of the suspect passage. After pilots on RAMDocs (Wang et al., 2025b), we pre-registered our hypotheses (H1–H9) and tested them on 300 held-out questions; eight addenda (H10–H31), each committed before its runs, extended the study to ConflictBank (Su et al., 2024) (including a variant where the misinformation outnumbers the truth), to RAGuard (Zeng et al., 2025), whose misleading side is real Reddit posts, and to seven generators from OpenAI, Anthropic and Qwen. Controls separate what a flag conveys: forced choice, an answer key, randomised direction, the detector's verdict and its model's own answer as hints, and six detectors crossed with six generators. WikiContradict (Hou et al., 2024), whose conflicts are genuine, is the contrast case.

Our findings:

- **A correct neutral flag does not reliably help.** Asked for every valid answer, OpenAI and Qwen models add the misinformation to their answer; asked for one answer, some generators abstain on misinformed and correct answers alike; under forced choice the effect is small. Claude

models move less, and the minimal prompt is the most harmful policy. These policy and family splits were coded after seeing data.

- **An oracle's direction is a ceiling.** Naming the false side helps on every benchmark that labels one, but it is equivalent to an answer key that names the wrong answer (pooled over 9 comparisons).
- **A directional flag pays only above a break-even precision.** When direction is randomised (H31), a wrong flag costs more than a right one gains, because generators follow both at similar rates and an accurate generator has more correct answers to lose. The break-even direction precision rises from 0.52 for gpt-4o-mini to 0.79 for gpt-6.1-terra, and the follow rates predict the effect of six real detectors on six generators (exploratory; Figure 1). The detector's verdict as an answer key matches its flag (H30); the detector model's own answer as a hint does worse (H27).
- **Direction is unsafe on genuine conflicts and costly when conflicts are rare.** On WikiContradict the LLM directional flag and removal cut the share of answers giving both valid sides (H28), and at a 5% share of conflict questions the real detector's flag lowers accuracy in 22 of 31 settings.

On CONFLICTS, the one benchmark of real search results, no condition changes accuracy significantly. We release the harness, pre-registrations and every run.[1]

2 Related Work

Knowledge conflicts. Language models can prefer memorised knowledge over contradicting context (Longpre et al., 2021), yet are readily persuaded by coherent contradicting evidence (Xie et al., 2024), to a degree that depends on the strength of their prior (Wu et al., 2024); Huang et al. (2025) train models to decide which of context and memory to trust, including on real incorrect Reddit contexts, and context-faithful prompting changes how readily models follow or abstain on conflicting context (Zhou et al., 2023). Of the conflict types of Xu et al. (2024), we study inter-context conflicts, as in Chen et al. (2022), who find that models tend to rely on a single passage when evidence conflicts, and Wan et al. (2024), who find that models weigh a passage's relevance more than its credibility cues.

Benchmarks. RAMDocs (Wang et al., 2025b), built on AmbigDocs (Lee et al., 2024), mixes genuine ambiguity, misinformation and noise. CONFLICTS (Cattan et al., 2025) annotates search results with conflict types, each with its own desired behaviour, and WikiContradict (Hou et al., 2024) pairs genuinely contradicting Wikipedia passages. FaithEval (Ming et al., 2025), ConflictBank (Su et al., 2024), ConfRAG (Yuan et al., 2026), QACC (Liu et al., 2025) and WhoQA (Pham et al., 2024) cover related settings. RAMDocs and ConflictBank label which side is misinformation; WikiContradict does not.

Telling the model about conflicts. Prompting a model to reason about the conflict type helps, and supplying the gold type helps much more (Cattan et al., 2025); on WikiContradict, models acknowledge both sides more often when told the passages may conflict (Hou et al., 2024). Ragability (Gross et al., 2026) finds that such a hint helps on contradictory contexts but hurts on *consistent* ones, where it is false. In our neutral-flag result the hint is true and the input contradictory, and accuracy still falls, because the hint does not say which side to trust. Pan et al. (2023b) test detection and "vigilant" prompting against LLM-generated misinformation, and Pan et al. (2023a) call for misinformation-aware QA. Source metadata and credibility signals also shape how models resolve conflicts (Chiang and Lee, 2024; Ge et al., 2025).

Telling the model which passage is wrong. Closest to our directional format, Hong et al. (2024) pass a discriminator's marks on perturbed passages to the generator in the prompt, which makes it more robust to counterfactual noise. CrAM (Deng et al., 2025) rescales attention by passage credibility, RA-RAG (Hwang et al., 2025) estimates source reliability, and Weller et al. (2024) defend open-domain QA against disinformation. TACS (Yu et al., 2024) detects untruthful context and masks it, close to our removal format. FaithfulRAG (Zhang et al., 2025) models conflicts at the level of facts before generating. These works, with Pan et al. (2023b)'s finding that vigilant prompts give limited protection, already show that a reliability signal can help more

[1] Code, the pre-registration and all run outputs are in the anonymised supplementary material and will be released publicly on acceptance.

than a warning, and some vary the signal's quality (Hong et al., 2024; Pan et al., 2024). What we add is a randomised manipulation of direction on the same pairs, answer-key controls that separate the verdict from the marking, its cost on conflict-free questions, and a break-even account of when a real detector's verdict helps a given generator. RGB (Chen et al., 2024) warns models that passages may contain errors, and they still rarely detect them; our neutral flag is the pair-specific version of that warning. Models also follow injected hints and asserted views, wrong ones included (Turpin et al., 2023; Sharma et al., 2024), and advice improves a decision-maker only when its errors fall where the decision-maker is right less often (Bansal et al., 2021; Mozannar and Sontag, 2020), which is the condition our break-even makes quantitative. Whether abstaining is a failure depends on the cost of a wrong answer (Wen et al., 2025; Song et al., 2025).

Detection, resolution and poisoning. ConflictRAG (Wang et al., 2026) trains a pairwise detector with an LLM fallback and reports pair-level F1 alongside answer gains; its NLI and chain-of-thought baselines resemble two of our detectors. LLMs detect conflicts but struggle to pinpoint them (Gokul et al., 2025; Wang et al., 2024; Jiayang et al., 2024), favour the more frequent side, the one matching their prior and LLM-generated passages (Jin et al., 2024; Kortukov et al., 2024; Tan et al., 2024), and are distracted by irrelevant context (Shi et al., 2023; Schuster et al., 2026). TCR (Ye et al., 2026), context-aware decoding (Shi et al., 2024), EvidentialRAG (Hossain et al., 2026) and credibility-aware generation (Pan et al., 2024) learn or decode around unreliable passages, and MADAM-RAG (Wang et al., 2025b), Astute RAG (Wang et al., 2025a), Self-RAG (Asai et al., 2024), Corrective RAG (Yan et al., 2024) and InstructRAG (Wei et al., 2025) handle them without an explicit contradiction detector. A few injected passages can steer a RAG system (Zou et al., 2025); RobustRAG (Xiang et al., 2024) isolates passages for certifiable robustness, and TrustRAG (Zhou et al., 2025) filters suspected poisoned ones, the analogue of our removal. We ask instead how a detector's output should be passed to the generator.

3 Framework

A question q comes with retrieved passages $P = (p_0, \ldots, p_{n-1})$. A *detector* returns flagged pairs $F \subseteq \{(i, j) : i < j\}$; a *directional* detector also returns, for each flagged pair, a *suspect* set $S_{ij} \subseteq \{i, j\}$: the side it believes is misinformation. A *format* turns (P, F, S) into the generator's prompt. We vary detectors and formats independently. Every prompt shares a skeleton: passages, question, and an instruction to end with a `Final answer(s):` line. The *default* skeleton asks for every valid answer when the passages support several (RAMDocs contains same-name ambiguity, so a correct answer may need several entities). The *minimal* skeleton drops that policy, and its answer line allows several answers without inviting them. The *single* skeleton says the question has exactly one correct answer (true of ConflictBank by construction). All three allow "unknown" when no passage answers the question; the *forced* skeleton is *single* without that option (Appendix A).

3.1 Formats

No flags (plain). Passages and question only.

Instruction. Plain plus a generic instruction to check the passages for contradictions, with no detector. v1 ends "if a disagreement reflects genuinely different valid answers, report each one"; v2 asks the model to judge which passages are reliable and to leave out answers supported only by unreliable ones.

Flag. Each flagged passage gets the header `[POTENTIAL CONFLICT with passage(s): j]`, followed by a list `Passages i and j: possible contradiction`. This is the original CAAG format, except that every detector's flags are rendered identically, so detectors differ only in *which* pairs they flag.

One-wrong flag. As Flag, but adds that exactly one passage of the pair contains misinformation, without saying which: it says that the conflict is misinformation, but not its direction.

Directional flag. As Flag, but suspect passages are headed `[LIKELY MISINFORMATION; conflicts with passage(s): j]`, and the list says which passage likely contains misinformation.

Controls. *Pick*: the neutral flag plus "decide which side is better supported and answer with

that side only". *No pairs*: each flag without its pair structure (flagged passages are marked, but not which conflicts with which). *Answer key*: plain plus "The answer X is false" for each wrong answer, with no passage marked.

Remove (suspect). The suspect passages of every flagged pair are deleted. For a pair without a suspect, the second passage is deleted, as the original CAAG REMOVE strategy does without retrieval scores.

Single call. No detector: one call that first writes `Suspect passages: ...` and then answers from the other passages, at half the calls of a detector pipeline.

Closed book. No passages; it measures what the generator knows without retrieval.

3.2 Detectors

CAAG (Contradiction-Aware Attention Gating) is a heuristic detector from an earlier, unpublished RAG system, used with its code unchanged. It scores similar pairs by negation, entity-mismatch and temporal cues, runs locally and gives no direction (Appendix B).

NLI is `nli-deberta-v3-base` (He et al., 2023), a cross-encoder scored in both directions; a pair is flagged if the larger contradiction probability exceeds 0.5. It gives no direction.

LLM is a single call that lists contradicting pairs, telling them apart from ambiguity, complementary and temporal relations, and names the side more likely to be misinformation (or "both" or "unclear"), as JSON. It is gpt-4o-mini (temperature 0) unless stated; in the crossed design gpt-4.1, gpt-6-luna, gpt-6.1-terra, Claude Haiku 5.5 and Claude Sonnet 5.5 run the same prompt.

Random flags each pair independently at CAAG's per-pair flag rate, a placebo that matches CAAG's flag volume.

Oracle flags exactly the gold contradiction pairs and names the gold misinformation side. The *inverted oracle*, an adversarial control, flags the same pairs but names the correct side.

3.3 Synthetic detectors

To separate a detector's accuracy from its identity, we build detectors from the gold labels with a chosen pair-level precision $p \in \{0.25, 0.5, 0.75, 1\}$, recall $r \in \{0.25, 0.5, 1\}$ and direction accuracy $d \in \{0.5, 0.75, 1\}$ (36 cells), each rendered as a directional flag and as removal. All cells reuse the same random draws, so they are nested and comparisons across cells are paired (Appendix E).

4 Experimental Setup

Data. **RAMDocs** (Wang et al., 2025b) has 500 questions, each with 1–12 documents (5.5 on average) typed *correct* (supports a gold answer, possibly for a different same-name entity), *misinfo* (supports a wrong answer) or *noise*. A pair is a *contradiction* if one side is misinformation and the answers differ, *ambiguity* if both are correct with different answers, *consistent* if the answers match, and *unrelated* if either is noise. A question is a *conflict* question if it has a contradiction pair, and *clean* otherwise. Some documents typed *correct* contain a wrong answer; questions whose only sign of conflict is such a document count only in "All" (Appendix D).

ConflictBank (Su et al., 2024): for a Wikidata fact, an LLM writes one evidence passage for the true object and one for a substituted object. We sample 400 questions round-robin over relations: 300 conflict questions (correct passage, misinformation passage, two distractors from other subjects) and 100 clean ones (correct passage, three distractors). Every question has one gold answer. It had no pilot, so all its results are confirmatory. A *majority* variant gives each conflict question the correct passage and ConflictBank's three wrong-object passages for that fact (its misinformation, temporal and semantic conflicts, all supporting the substituted object, which we count as misinformation), so the false side outnumbers the truth three to one.

RAGuard (Zeng et al., 2025) pairs PolitiFact claims (true or false) with Reddit posts that an LLM labelled supporting, misleading or unrelated, so the misleading side is real online text. We keep 294 claims with at least one supporting and one misleading post (up to two of each, plus two unrelated posts) and 100 claims with no misleading post; the question is whether the claim is true. Its labels are LLM-assigned stances, not verified facts.

WikiContradict (Hou et al., 2024) has 253 questions, each with two genuinely contradicting Wikipedia passages; the reference answer is *both* sides. We also run 237 automatically scorable **CONFLICTS** (Cattan et al., 2025) questions (Ap-

pendix D).

Pre-registration and splits. We developed the method on a random 200-question pilot split of RAMDocs. Before running anything on the remaining 300 questions (139 conflict, 126 clean, 35 excluded), we committed a pre-registration that fixes the conditions, metrics, hypotheses and corrections. Every RAMDocs result in the main text is on this confirmatory split; pilot results appear only in the appendix. Comparisons that were not registered, including every real-detector row of Table 5, are exploratory. Eight addenda registered H10–H31, each committed before its own runs, so these hypotheses were formed after the original confirmatory results were known; the confirmatory core registered before any data is H1–H9. The last two addenda also registered re-analyses of data we had already seen, and a final round of re-analyses (pooled answer-key comparisons, break-even precision, the prediction of the crossed cells, prevalence reweighting and a programme-wide FDR) was not registered at all; all are reported as exploratory (Table 2). The repository's commit history is the only timestamp. Appendix D lists every deviation.

Generators and baselines. The main generator is `gpt-4o-mini` (OpenAI et al., 2024) (temperature 0.1, at most 256 output tokens, seeds 0–2). We replicate key conditions with `gpt-4.1`, Qwen2.5-7B-Instruct (Qwen Team et al., 2024) (4-bit, local), two current reasoning models, `gpt-6-luna` (3 seeds) and `gpt-6.1-terra` (seed 0; one sample), and Anthropic's Claude Haiku 5.5 (3 seeds) and Claude Sonnet 5.5 (seed 0). The reasoning and Claude models run at their default temperature with up to 4,096 output tokens, since they reason before answering. The LLM detector is gpt-4o-mini except in the crossed design (H24), where six detectors and six generators form 30 pairs on ConflictBank. On ConflictBank the primary prompt is the single skeleton; the default and minimal skeletons are robustness checks. We rerun the oracle and LLM-detector conditions with the minimal skeleton. MADAM-RAG (Wang et al., 2025b) is ported unchanged (three debate rounds). Astute RAG (Wang et al., 2025a) and TrustRAG (Zhou et al., 2025) use their published prompts and defaults; since both return one answer, we also run *multi* variants that ask for every valid answer (Appendix D). We also run the LLM detector on Qwen2.5-7B with gpt-4o-mini generating. A detector pipeline counts as one detector call plus one generator call. Two controls test whether a crossed cell's gain is only a stronger model's answer (H27): the generator given the detector model's own plain answer as a hint, and the detector model used as the generator. An artefact probe (H29) reruns two detectors on ConflictBank with the passage order reversed and with every passage paraphrased by one model.

Metrics. On RAMDocs an answer is correct if it includes every gold answer and no wrong answer (Wang et al., 2025b). Our primary metric reads "includes" literally: after normalisation, each gold or wrong answer must appear as a whole-word span of the extracted final answer. A bidirectional variant, which also counts a predicted answer contained in a gold one, is the robustness metric. We chose the one-way metric before the confirmatory run because it is closer to MADAM-RAG's published score (Appendix D), and we compare conditions only within our own harness. On WikiContradict an answer is correct if it contains both sides (bidirectional matching); on ConflictBank and RAGuard we use the RAMDocs metrics. We also report the share of answers that contain the misinformation, the share that abstain ("unknown"), and accuracy among answered questions. An LLM judge (gpt-6.1-terra) agrees with the primary metric on 92.6% of 162 stratified conflict-question outputs ($\kappa = 0.85$); the metric is slightly conservative, missing some paraphrases. A 200-output sheet for a human audit is released, but it has not been labelled.

Statistics. We average each question's score over seeds and report paired differences from the plain format with bootstrap 95% CIs that resample questions (10,000 resamples) and two-sided sign-flip permutation tests (Efron and Tibshirani, 1993; Dror et al., 2018) (10,000 permutations, extended to 200,000 when none reaches the observed statistic). A *family* is the set of tests one hypothesis registers (e.g. H15 across generators); a registered test takes its family's Holm-adjusted p (Holm, 1979), the 636 exploratory comparisons of the condition tables form one family, and the hypothesis tables add a Benjamini–Hochberg q over all registered tests. Only results significant under Holm are called significant or set in bold; CIs are unadjusted and are not read as significance. A result is *robust* if it also holds under the bidirectional metric. A null difference is not read as equivalence without

a TOST (Lakens, 2017) at ±3 points. Selective accuracy is a ratio (correct over answered), tested by swapping conditions within questions. Pooled estimates use a multilevel model with random effects for generator, benchmark and setting. Appendix D gives the minimum detectable effect.

Cost. Token counts come from the API's usage reports. All experiments, pilots included, cost $187.38 in API fees (prices checked on 2026-10-27; Appendix D); Qwen ran locally.

5 Results

Tables 4 and 5 give every condition, the hypothesis tables every registered test, and Table 2 the status of each claim below: registered before any data (H1–H9), registered before its runs in a later addendum (H10–H31), or a re-analysis of data already seen (Appendix B).

5.1 What a neutral flag does depends on the answer policy

Flagging exactly the gold contradictions in the neutral format lowers gpt-4o-mini's RAMDocs conflict-question accuracy by 6.7 points [−10.8, −3.1] from 22.1% (H2), and gpt-4.1's by 8.6 (H9). How depends on what the prompt asks for (Table 1).

Asked for every valid answer, the model adds the misinformation. Under RAMDocs' default prompt, which asks for every valid answer, the neutral flag raises the share of gpt-4o-mini's answers that contain the misinformation from 47.7% to 61.9% (and on ConflictBank under the same prompt from 72.8% to 83.2%): told that two passages conflict, it reports both sides. Every OpenAI and Qwen generator moves the same way (e.g. gpt-4.1 on ConflictBank 82.7% to 90.7%); both Claude models move the other way (Haiku on RAMDocs 55.2% to 47.2%).

Asked for one answer, it abstains. With a prompt stating that the question has one answer, the neutral flag costs gpt-4o-mini 18.7 points on ConflictBank, gpt-4.1 8.0 and gpt-6-luna 9.4 (H15, H19), and changes accuracy insignificantly for gpt-6.1-terra, Qwen2.5-7B, either Claude model (H25) and every generator on RAGuard (H26). Of gpt-4o-mini's misinformed answers, 45% abstain with the flag and 7% become correct (3% between two unflagged seeds), and 37% of its correct answers also abstain; accuracy on the answers it still gives moves by −0.3 to +3.9 points (H21). When "unknown" is not allowed (H20), the flag has no significant effect for gpt-4o-mini (−0.2), gpt-6-luna (+1.6) or gpt-6.1-terra (−0.3), equivalence not established, and a small one for Claude Haiku (+3.9).

Across settings. In a multilevel model over the 34 settings (Table 37; exploratory), the pooled effect is −1.6 points [−5.5, +2.2] and a new setting in this design space would fall in [−13.6, +10.3]. It is −12.1 [−23.6, −0.7] under the minimal prompt, which states no answer policy, −3.8 under multi-answer prompts, −0.5 under one-answer prompts and +1.5 under forced choice; Claude (+5.2) shifts it up about as much as forced choice (+6.4). The policy and family codings were chosen after seeing data. If abstention is costless and a misinformed answer costs c wrong ones, the flag lowers utility significantly in 11 of 34 settings at $c = 0$, and raises it in 4 at $c = 4$ (Table 38). On WikiContradict, where both sides are valid, listing both is correct, and the flag raises accuracy (H8: +6.5).

5.2 Naming the false side: what an oracle can deliver

With the same oracle pairs, the directional flag beats the neutral and one-wrong flags in every registered comparison but one (H3, H9–H12, H15, H16, H19, H25, H26; gpt-6.1-terra on RAGuard is not significant). On ConflictBank it gains 17.0 to 36.4 points for the OpenAI generators and 16.3 to 23.3 for Claude, under forced choice (H20: +14.7 to +26.4), under majority misinformation (+32.7 to +66.3) and on RAGuard (H26: +6.8 to +20.4). On RAMDocs it gains 13.4 points with gpt-4o-mini and up to 39.6 with gpt-6.1-terra, but not significantly for Qwen2.5-7B or for gpt-4o-mini under the minimal prompt (H10). The ConflictBank ranges are for the one-answer prompt.

These rows are a ceiling, not a detector result. Telling the model "the answer X is false", with no passage marked, does as well: no registered comparison differs significantly (H23; −5.0 to +5.3 points), and pooled over the 9 comparisons, with a bootstrap that resamples each benchmark's questions jointly across them, the flag's advantage is −0.1 points [−1.5, +1.2], equivalent within ±3 (90% CI [−1.3, +1.0]; exploratory). An inverted oracle that names the correct side costs 7.0 points on RAMDocs (Table 3). Saying only that one side

Generator	No flags				Oracle, neutral flag				Oracle, directional flag			
	Correct	Misinfo.	Abstain	Sel.	Correct	Misinfo.	Abstain	Sel.	Correct	Misinfo.	Abstain	Sel.
RAMDocs, multi-answer prompt												
gpt-4o-mini	22.1	47.7	3.4	22.8	15.3	**61.9**	5.0	16.2	35.5	**14.9**	2.9	**36.5**
gpt-6-luna	14.4	72.2	1.2	14.6	11.3	76.7	0.7	11.4	53.5	**11.3**	1.4	**54.3**
Claude Haiku 5.5	25.2	55.2	2.2	25.7	26.4	**47.2**	2.2	27.0	53.5	**6.7**	2.9	**55.1**
gpt-6.1-terra	10.8	75.5	2.2	11.0	10.1	81.3	1.4	10.2	50.4	**13.7**	2.2	**51.5**
Claude Sonnet 5.5	45.3	24.5	2.2	46.3	45.3	21.6	2.2	46.3	59.7	**3.6**	1.4	**60.6**
ConflictBank, multi-answer prompt												
gpt-4o-mini	21.6	72.8	0.3	21.6	12.8	**83.2**	0.9	**12.9**	49.4	**41.2**	1.2	**50.1**
gpt-4.1	13.0	82.7	0.7	13.1	6.7	**90.7**	0.3	**6.7**	36.0	**58.3**	1.0	**36.4**
Qwen2.5-7B	23.7	67.0	6.0	25.2	15.3	**79.0**	5.0	**16.1**	41.3	**52.0**	13.0	**47.5**
ConflictBank, one-answer prompt												
gpt-4o-mini	51.1	31.0	6.7	54.8	32.4	**17.7**	**41.0**	55.0	81.2	**4.2**	**2.1**	**83.0**
gpt-6-luna	53.6	24.3	11.9	60.8	44.1	**17.4**	**28.9**	62.0	90.0	**0.0**	**0.8**	**90.7**
Claude Haiku 5.5	66.8	22.0	1.2	67.6	67.4	22.1	0.9	68.0	90.1	**0.2**	1.1	**91.1**
gpt-6.1-terra	73.0	18.0	1.7	74.2	70.7	15.7	4.7	74.1	90.0	**0.7**	0.3	**90.3**
Claude Sonnet 5.5	75.7	15.7	1.0	76.4	78.7	12.0	1.7	80.0	92.0	1.0	1.7	93.6
gpt-4.1	64.7	27.3	1.7	65.8	56.7	21.7	14.0	65.9	91.3	0.3	1.7	92.9
Qwen2.5-7B	50.3	35.3	5.0	53.0	47.0	28.7	15.0	55.3	75.7	13.0	1.3	76.7
ConflictBank, forced choice												
gpt-4o-mini	55.4	31.6	0.1	55.5	55.2	31.9	0.4	55.5	79.7	**6.7**	0.0	**79.7**
gpt-6-luna	64.6	25.8	0.0	64.6	66.1	22.0	0.0	66.1	91.0	**0.1**	0.0	**91.0**
Claude Haiku 5.5	69.3	20.9	0.0	69.3	73.2	**16.8**	0.0	**73.2**	90.4	**0.0**	0.0	**90.4**
gpt-6.1-terra	77.7	14.7	0.0	77.7	77.3	13.3	0.0	77.3	92.3	**0.3**	0.0	**92.3**

Table 1: Risk and coverage on conflict questions (%): strictly correct answers, answers that contain the misinformation, abstentions ("unknown"), and selective accuracy (correct answers over non-abstained answers, pooled over questions and seeds). **Bold**: the change from no flags is significant (H21 for ConflictBank one-answer and forced; the paper's exploratory family elsewhere). The answer policy decides what a neutral flag does: under a prompt that asks for every valid answer it adds the misinformation to the answer; under a one-answer prompt that allows "unknown" it moves answers, misinformed and correct, into abstention; under forced choice it has no abstention to trade.

is false helps when the model must answer: the one-wrong flag lowers gpt-4o-mini's one-answer ConflictBank accuracy by 35.7 points (H16) but raises every generator's under forced choice (H20: +5.1 to +10.3).

5.3 With a real detector, its wrong flags decide

Registered results. With the gpt-4o-mini LLM detector, the directional flag significantly raises accuracy over all ConflictBank questions for 1 of 7 generators (gpt-6-luna, +6.1) and significantly lowers it for 1 (gpt-6.1-terra, −8.8; H15, H19, H25). It helps all 4 generators under majority misinformation (+5.7 to +9.0), and on RAGuard the two Claude models (H26: +5.0, +7.1) but no OpenAI one.

A wrong flag costs more than a right one gains. In a randomised test (H31), every ConflictBank conflict question got a flag on its true contradiction, naming the false side on a random 159 questions and the correct side on the other 141. A right flag raises accuracy by +17.6 to +33.5 points and a wrong one lowers it by 31.2 to 64.5 (Figure 1, left). At this split the net is significantly negative for 3 of 4 generators. Each generator therefore has a break-even direction precision, $-W/(R-W)$ for right- and wrong-flag effects R and W, below which a flag on the true pair does net harm: 0.52 [0.43, 0.60] for gpt-4o-mini, 0.57 for gpt-6-luna, 0.67 for Claude Haiku and 0.79 [0.72, 0.85] for gpt-6.1-terra (Table 24; exploratory). Generators follow right and wrong flags at similar rates (e.g. gpt-4o-mini 57% [46, 67] and 51% [40, 62]), except gpt-6.1-terra (67% and 95%, one seed), so the break-even rises mainly because an accurate generator has more correct answers to lose.

The randomised test predicts real detectors. H24 crossed six detectors with six generators in 30 cells. For each cell we predict the flag's effect from the cell's own detections and no-flag answers and the follow rates measured in H31's separate runs: a flag naming the false side on a question the generator gets wrong gains its right-follow

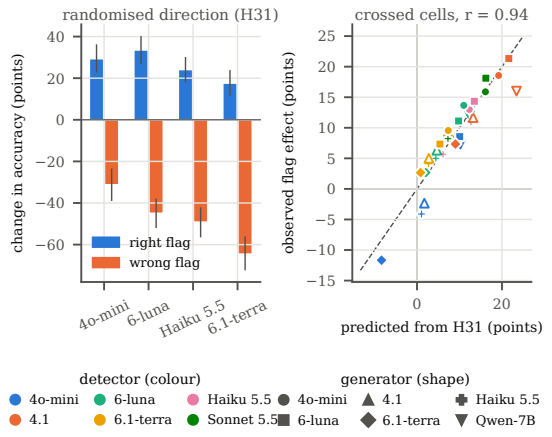


Figure 1: Left: randomised direction (H31; ConflictBank, one-answer prompt). Change in accuracy on the conflict questions whose flag named the false side (right) and on those whose flag named the correct side (wrong), with 95% CIs. Right: the 30 crossed detector–generator cells; the flag's effect predicted from H31's follow rates against the observed effect (filled: the generator's own follow rates; open: the mean follow rates, for generators with no H31 run); dashed: $y = x$. Exploratory. Table 35 lists the cells.

rate, and one marking a correct passage on a question it gets right costs its wrong-follow rate. For the 22 cells whose generator had an H31 run, prediction and observation agree closely ($r = 0.97$ [0.93, 0.99], slope 1.06, intercept -0.54, mean error 1.5 points; within generators $r = 0.93$), and the prediction calls the harm the gpt-4o-mini detector does to gpt-6.1-terra. For the 8 cells of generators without one, with the mean follow rates, $r = 0.90$ but the prediction overshoots (slope 0.70; Figure 1, right). This prediction was defined after the cells were seen. The registered H24 correlation ($\rho = +0.64$, $p<0.001$) is uninformative: the generator's accuracy enters both sides, and leaving out one detector it predicts held-out cells worse than their mean (R^2 =-0.21). Within generators, direction precision and recall both matter ($p = 0.047$ and 0.040, exact permutation over the six detectors; exploratory). The gpt-4o-mini detector, the least precise (68.8% against 77.7–82.6%), helps the weaker generators and lowers the accuracy of the four most accurate, significantly for gpt-6.1-terra (H19).

What the detector contributes is its verdict. Passing the detector's judgement as an answer key ("the answer X is false" for the answer its suspect passage supports, no passage marked) matches the directional flag (H30: -2.0 to $+3.3$ points per cell, none significant; pooled over the 22 cells the flag is ahead by $+1.2$ [-0.1, $+2.5$], equivalent within ± 3) and beats the detector model's own answer as a hint by $+5.3$ [$+3.2$, $+7.6$]. That hint does worse than the flag in 14 of 30 cells and never better (H27).

When conflicts are rare. The LLM detector also flags clean questions (76.2% of RAMDocs'), where a directional flag can only hurt; at a 5% share of conflict questions its flag lowers accuracy in 22 of 31 settings (Table 25; exploratory).

Artefacts, genuine conflicts and baselines. Reversing the passage order (Liu et al., 2024) or paraphrasing every passage changes the detectors' hit rate by -1.7 to $+1.0$ points (H29), none significant, with equivalence within ± 5 shown for one of four. Where both sides are valid, direction does damage (H28): on WikiContradict the LLM directional flag lowers the share of answers giving both sides by 6.5 points for gpt-4o-mini, 29.0 for gpt-6-luna and 27.7 for Claude Haiku, and on the 99 RAMDocs same-name questions removal costs every generator 21.2 to 29.3 points. In the registered ConflictBank comparison with Astute RAG and TrustRAG our pipeline does not win (H14; Appendix B).

6 Discussion

A neutral flag does what the answer policy lets it. It says that passages disagree but not which to trust, so the model falls back on its policy and its family's habits. Abstaining is right where a wrong answer is costly (Wen et al., 2025), so a neutral flag is worth sending only with a one-answer policy and a measured cost for wrong answers.

A directional flag is advice, and its value has a threshold. Generators act on wrong verdicts about as readily as on right ones, as with sycophancy (Sharma et al., 2024), so the flag helps only above a break-even precision that rises with the generator's accuracy: the complementarity condition of human–AI teams and of learning to defer (Bansal et al., 2021; Mozannar and Sontag, 2020). This suggests, though no gated pipeline was tested end to end, estimating a generator's break-even with a randomised-direction test and a detector's direction precision on the target distribution, clean questions included, before passing direction.

Limitations

Models. We test seven generators from three families: four OpenAI models (gpt-4o-mini, gpt-4.1 and two current reasoning models), Claude Haiku 5.5 and Sonnet 5.5, and Qwen2.5-7B at 4-bit precision. We could not run Gemini, and no large open model. Qwen2.5-7B, gpt-6.1-terra and Claude Sonnet ran with one seed, as did gpt-4.1 outside RAMDocs, and the reasoning and Claude models only at their default temperature, so their sampling is not deterministic, and Claude ran through Anthropic's OpenAI-compatible endpoint. The crossed reliability test (H24) covers 30 detector–generator cells on one benchmark, with six detectors, so any test of a detector-level property has little power (an exact permutation has 720 arrangements); the registered gap turned out to be dominated by the generator's own accuracy. The prediction from H31's follow rates was defined after seeing the cells, uses follow rates measured with oracle pairs on the same questions, and is validated out of sample only on the 8 cells of generators without an H31 run. Concurrent runs gave a detector slightly different outputs in different cells; its output is identical across its cells for at least 90% of questions, and restricting each cell to those moves no cell's gain by more than 1.3 points. The LLM detector is a single prompt. The published baselines are reimplemented (Astute RAG has no official code), and we do not compare against a trained detector such as ConflictRAG's (Wang et al., 2026).

Data and power. The confirmatory evidence comes from two benchmarks: RAMDocs (300 questions, 139 of them conflict questions) and ConflictBank (400 sampled questions, 300 with conflicts). In both, the misinformation is synthetic, made by editing or generating text around a substituted answer; ConflictBank's passages are LLM-written on both sides. RAGuard's misleading side is real, but its stance labels are assigned by an LLM. With its posts in context generators abstain on 32.1–60.9% of conflict claims, which is why accuracy on this true/false task falls below chance; on the claims they answer it is 53.8–68.4%, against 64.1–74.1% closed-book. RAMDocs' document labels are imperfect. We exclude 35 hold-out questions whose status depends on documents we could not label without human annotation, and we check that the conclusions survive treating those documents as misinformation. Our comparisons detect effects of about 5.8 points with 80% power, so smaller effects, including small harms on clean questions, may be missed. WikiContradict has genuine conflicts but no misinformation side. CONFLICTS shows no significant effect for any condition; scored only on items with a reference answer, it can neither confirm nor refute our findings. The majority variant tests one case where the false side is more numerous, but its extra wrong-side passages are generated for the same substitution, which a real attacker would vary, and they include ConflictBank's temporal and semantic conflicts, which we count as misinformation. On RAGuard every generator is more accurate closed-book than with the posts (gpt-4o-mini 64.9% against 36.7%), so its posts make a weak retrieval setting. The current reasoning models distrust ConflictBank's generated passages when asked to judge them (Appendix C), which suggests that generated benchmarks may not measure how such models treat real documents. Reversing and paraphrasing the passages change the detectors' hit rate insignificantly (H29), but equivalence within ± 5 points is shown for only one of four changes, so position and surface style are not ruled out as sources of their direction, nor are deeper regularities of generated text.

Metric. RAMDocs ships no scorer. Our primary metric reconstructs the published definition and scores our MADAM-RAG port close to the published number on the pilot split, but the published subset is unknown, so this is not a reproduction. The metric is string matching: it misses paraphrases, and partial answers interact with "no wrong answer". An LLM judge agrees with it on 92.6% of sampled outputs, but no human labels check either; the 200-output audit sheet we release is unlabelled, and the judge, gpt-6.1-terra, is itself one of the generators. We report a more lenient variant throughout and mark any conclusion that depends on the choice.

Direction and the grid. Oracle direction is not distinguishable from an explicit answer key (H23), so the oracle rows are ceilings, not findings about detectors. In the synthetic grid the gain rises with direction accuracy, but the grid cannot separate adjacent levels. The gpt-4o-mini LLM detector's direction precision (of its flags on a true contradiction, the share naming the false side; distinct from its pair precision, the share of its flagged pairs that are true contradictions) is 88.3% on RAMDocs

and 68.8% on ConflictBank; its hit rate per question is lower, because it also misses pairs. Its directional flag's gain over all ConflictBank questions with gpt-4o-mini is not significant after correction (H15), and with stronger generators it hurts. Synthetic false flags are spread uniformly and their suspect side is random, whereas real detectors concentrate their errors. The grid predicts real detectors only roughly, and three detectors cannot show whether document- or pair-level summaries predict better.

Pre-registration. The original hypotheses (H1–H9) were registered before the confirmatory run. H10–H31 were registered in eight addenda, after the earlier results were known though before their own runs, and the only timestamp is the repository's commit history, not a third-party registry. Holm controls error within each family, not across the programme: of 518 registered tests, 266 are significant under Holm and 294 under a Benjamini–Hochberg FDR over all of them (both shown in the hypothesis tables). Several analyses were chosen after seeing data and are labelled as such: the TOST margin, the single- and multi-answer split, the document-level placement axes, the post-hoc comparison of detector and generator reliability that H24 later tested, H24's mechanism score, and the refinement of permutation p-values at their floor.

Scope. We study flags passed to the generator as text. Learned interfaces such as soft prompts or credibility fine-tuning (Ye et al., 2026; Pan et al., 2024), and retrieval-side filtering, may follow different trade-offs. We test one adversarial case, a detector that always names the wrong side, but not an attacker who controls which pairs are flagged.

Ethics Statement

This work uses public benchmarks and commercial model APIs, and involves no human subjects or personal data. Its main risk is dual use. Our results show that a false "likely misinformation" label, or the removal of a passage, can change a RAG system's answers: an inverted oracle that names the correct side as misinformation costs 7.0 points on RAMDocs and 8.6 on ConflictBank as a flag, and 21.8 on RAMDocs when the passage it names is removed. An adversary who can trigger a deployed detector could exploit this (Zou et al., 2025). We therefore report harm on conflict-free questions alongside every gain, and we argue that systems should measure what a detector does at the document level before acting on its flags.

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A Prompts

All prompts below are copied verbatim from the code that produced the results.

Generation skeleton (default).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Minimal skeleton (no answer policy).

```
Answer the question using the passages below.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your answer; if there is more than
one, separate them with |>
```

Single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Closed book (single-answer version).

```
Answer the question from your own knowledge. The
question has exactly one correct answer.
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Instruction v1.

```
Some passages may contradict each other. Before
answering, check whether they do.
Ignore passages that appear to contain
misinformation, and if a disagreement reflects
genuinely different valid answers, report each one.
```

Instruction v2.

```
Some passages may contain false information. Before
answering, decide which passages are reliable by
checking them against each other.
Do not include an answer that is supported only by a
passage you judge to be unreliable.
```

Single call (inserted as the instruction).

```
Some passages may contain misinformation. First
decide which passages, if any, contain
misinformation, by checking them against each other
and against what you know, and write one line in
this format:
Suspect passages: <passage numbers, or none>
Then answer the question using only the other
passages.
```

Example: directional flag (passage 2 suspect).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".
```

```
Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [LIKELY MISINFORMATION; conflicts with
passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; passage
2 likely contains misinformation

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Example: one-wrong flag, single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [CONFLICT with passage(s): 2; one side
contains misinformation]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [CONFLICT with passage(s): 0; one side
contains misinformation]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; exactly
one of the two contains misinformation (it is not
known which)

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Directional LLM detector.

```
You will see a question and {n} numbered passages
retrieved for it.
Some passages may contain misinformation: false
claims about the answer to the question.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

For each contradiction, say which passage of the
pair is more likely the misinformation, judging
by agreement with the other passages and by
plausibility: its number, "both" if both are likely
false, or "unclear" if you cannot tell.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
<int>, "relation":
"contradiction|ambiguity|complementary|temporal",
"misinformation": <int>|"both"|"unclear"}}]}}
Use the passage numbers shown. "misinformation" is
needed only for contradictions. Return {{"pairs":
[]}} if there are none.
```

LLM detector without direction.

```
You will see a question and {n} numbered passages
retrieved for it.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
<int>, "relation": "contradiction|ambiguity|compleme
ntary|temporal"}}]}}
Use the passage numbers shown. Return {{"pairs":
[]}} if there are none.
```

Forced-choice skeleton (no "unknown").

```
Answer the question using the passages below. The
question has exactly one correct answer.
You must give your best answer; "unknown" is not
allowed.
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Pick instruction (with the neutral flag).

```
When flagged passages conflict, decide which side is
better supported by the other passages and by what
you know, and answer with that side only.
```

Example: neutral flag, single-answer skeleton.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [POTENTIAL CONFLICT with passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction
```

```
Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: neutral flag without pairs.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [POTENTIAL CONFLICT]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
Some of the marked passages conflict with each
other.

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: one-wrong flag without pairs.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [IN CONFLICT; some marked passages contain
misinformation]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [IN CONFLICT; some marked passages contain
misinformation]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
Some of the marked passages contain misinformation
(it is not known which).

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: directional flag without pairs.

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [LIKELY MISINFORMATION]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
Passages marked LIKELY MISINFORMATION probably
contain false information.

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: answer key (also the verdict key, with the detector's answers).

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0:
<passage 0>

Passage 1:
<passage 1>

NOTE: The answer "<wrong answer>" is false.

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: hint (the detector model's own answer).

```
Answer the question using the passages below. The
question has exactly one correct answer.
If no passage answers the question, the answer is
"unknown".

Passage 0:
<passage 0>

Passage 1:
<passage 1>

NOTE: Another model that read these passages
answered: "<other model's answer>".

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your one answer>
```

Example: WikiContradict (default skeleton, neutral flag).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 1]:
<passage 0>

Passage 1 [POTENTIAL CONFLICT with passage(s): 0]:
<passage 1>
```

```
POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 1: possible contradiction

Question: <question>

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Paraphrase prompt (artefact probe, H29).

```
Paraphrase the passage below. Keep every fact, name,
number, date and place exactly as stated, including
any that seem wrong; do not add, remove or correct
information, and keep about the same length. Output
only the paraphrased passage.

Passage:
{passage}
```

LLM judge of the string matcher.

```
You are checking how a model's answer was scored.
Read the question, the correct answer(s), the wrong
(misinformation) answer(s), and the model's full
response. Judge only the model's final answer.

Question: {question}
Correct answer(s): {gold}
Wrong answer(s): {wrong}

Model response:
<<<
{response}
>>>

Reply with JSON only: {{"gives_all_correct":
true/false, "gives_wrong": true/false, "abstains":
true/false}}
- gives_all_correct: the final answer states every
correct answer (paraphrases and aliases count).
- gives_wrong: the final answer states any wrong
answer as an answer (mentioning it only to reject it
does not count).
- abstains: the final answer is "unknown" or
declines to answer.
```

B Additional Results

Published baselines (H14). Astute RAG (Wang et al., 2025a) and TrustRAG (Zhou et al., 2025) lose same-name answers on RAMDocs (H14). On ConflictBank the registered H14 compared our pipeline under the multi-answer prompt with the baselines' own one-answer prompts; under the matched one-answer prompt it beats them by +9.2 and +13.4 points (exploratory). A single call that names suspect passages before answering does not differ significantly from the pipeline (H18; Table 34).

Robustness to the metric and the labels. Table 28 repeats Table 5 with the bidirectional metric. Table 29 repeats the pre-registered tests with every suspect document treated as misinformation. Table 6 lists all pre-registered tests.

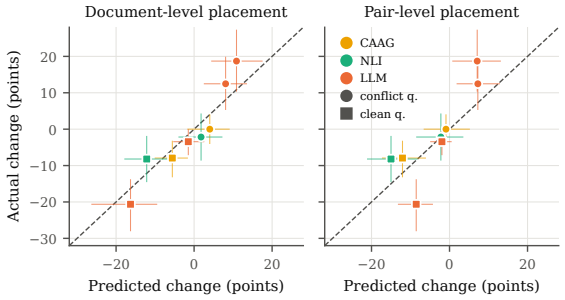


Figure 2: Effect of real detectors predicted from the synthetic grid (x, with the 95% bootstrap prediction interval) against the measured effect (y, with its 95% CI), for conflict (●) and clean (■) questions. Left: detectors located by their document-level behaviour; right: by pair precision, recall and direction accuracy.

The heuristic detector. CAAG scores a pair only if its MiniLM (Reimers and Gurevych, 2019) cosine similarity reaches $\theta_s = 0.65$. Its score is the sum of a negation and contrast asymmetry term, a named-entity mismatch term (differing dates, numbers, money or percentages) and a temporal-marker term, multiplied by the similarity. A pair is flagged if the score reaches $\theta_c = 0.35$. Its per-pair flag rate on the confirmatory split (6.22%) sets the rate of the random detector. CAAG is the detector we set out to study. As a neutral flag it did not beat random flags of the same volume, and as a removal rule it only deleted correct passages (−7.9 points on clean questions, 0.0 on conflict ones). Its false flags fall mostly on pairs that should both be reported: 134 of its 221 false flags on the hold-out are same-name ambiguity pairs, which its entity-mismatch signal flags because they mention different dates or numbers. We read this as a warning about lexical contradiction cues in general (Gururangan et al., 2018), rather than as a problem with a particular threshold. Retuning illustrates why F1 misleads. Over a 121-setting sweep, CAAG's best pair F1 is 29.2, above NLI's, at $\theta_s = 0.40$ and $\theta_c = 0.10$. But it gets there by flagging 43% of all pairs and 87% of clean questions, which our grid predicts would be harmful under removal (Appendix B).

Original CAAG format. Rendering CAAG's flags in its original format, with typed lines and confidence scores ("factual conflict (confidence: 0.41)"), changes accuracy by −0.9 points overall (−1.2 on conflict questions, −1.6 on clean ones). This is no different from the uniform format.

Claim (section)	Evidence	Status
A correct neutral flag lowers RAMDocs accuracy (§5.1)	H2, H9	registered before any data
Its mechanism depends on the answer policy: added misinformation, abstention, little effect (§5.1)	Table 1; H15, H19–H21, H25, H26	answer mix: re-analysis; tests: registered addenda
Pooled effect, policy split, moderators, utility, transitions (§5.1)	Tables 37, 38	re-analysis of data already seen
The directional flag beats the neutral one (§5.2)	H3, H9–H12, H15, H16, H19, H25, H26	H3: before any data; rest: addenda
Oracle direction is close to an answer key (§5.2)	H23 and its TOST	registered addendum; TOST registered in §13.4
Pooled, oracle direction is equivalent to an answer key (§5.2)	pooled H23	unregistered re-analysis
Real-detector gains depend on the generator (§5.3)	H15, H17, H19, H25, H26	registered addenda
Right flags gain less than wrong flags cost; similar follow rates (§5.3)	H31	registered addendum (§16)
Break-even direction precision rises with generator accuracy (§5.3)	Table 24	unregistered re-analysis
H31's follow rates predict the crossed cells (§5.3)	Figure 1	unregistered re-analysis
Registered reliability gap correlates with the gain (§5.3)	H24	registered addendum (§13), 5 of 30 cells existed
The gap is dominated by generator accuracy; precision and recall matter within generators (§5.3)	Table 36	re-analysis of data already seen
The detector's verdict as an answer key matches the flag; its model's answer as a hint does not (§5.3)	H30, H27	registered addenda (§15, §16); pooled TOST unregistered
A real detector's flag hurts when conflicts are rare (§5.3)	Table 25	unregistered re-analysis
Direction survives reordering and paraphrase (§5.3)	H29	registered addendum; equivalence not shown
Direction and removal harm genuine conflicts (§5.3)	H28	registered addendum
Recommendations (§6)	–	not tested end to end

Table 2: The status of each claim. *Registered addendum*: committed to the repository before the runs it covers but after earlier results were known. *Re-analysis*: registered after the data were seen, or exploratory. The commit history is the only timestamp.

Inverted oracle (names the correct side)	from no flags	from 50 items
RAMDocs, directional flag	−7.0 [−11.5, −2.9]	0.0
RAMDocs, removal	−21.8 [−28.8, −15.3]	0.0
ConflictBank (multi-answer), directional flag	−8.6	0.0

Table 3: The adversarial control, gpt-4o-mini, change from no flags in points (exploratory; §9.5). On clean questions it flags nothing, so the prompt is unchanged.

Pilot study (exploratory). On the 200-question pilot split, scored with the bidirectional metric, the same pattern appeared before any confirmatory run. Oracle neutral flags changed conflict-question accuracy by −10.5 points, directional flags by +11.8, and removing the suspect by +28.4. The pilots also showed that the original REMOVE strategy, which deletes an arbitrary side of each pair, removes correct documents. They are why we added directional formats, synthetic direction accuracy and document-level placement. All pilot runs are released.

Phase-0 audit of CAAG on NarrativeQA. CAAG was designed for long-document QA. On 100 NarrativeQA (Kočiský et al., 2018) questions from 50 stories (BM25 top-10 chunks from a single coherent story), it flags at least one pair in 54% of questions, 0.0166 pairs in all. A single story rarely contradicts itself, so most of these flags are likely false; the flagged pairs are released for annotation. This is the setting where a false flag can only hurt.

C Further Results

C.1 Where the directional flag fails (exploratory)

After seeing the minimal-skeleton run, we split conflict questions by the number of gold answers. 105 of 139 have several (same-name ambiguity *and* misinformation), and a strict answer must list all of them. On the 34 single-answer conflict questions, the directional flag's estimate is positive in every setting, from +11.8 points (Qwen) to +46.1 (gpt-4o-mini), both with the default skeleton. On multi-answer questions it changes little except with gpt-4.1 (gpt-4o-mini +2.9, and −5.7 with the minimal skeleton; gpt-4.1 +19.0). One reading is that a warning about one passage makes the smaller models more conservative about the legitimate alternative answers too; we did not test

			Accuracy		Oracle pairs				LLM detector		Single	LLM dir.
Generator	Prompt	Seeds	Closed	Plain	Flag	One wrong	Dir. flag	Remove	Flag	Dir. flag	call	clean q.
gpt-4o-mini	single	3	10.0	51.1	**−18.7**	**−35.7**	**+30.1**	**+35.0**	**−18.2**	+8.4	+7.6	−4.7
gpt-4o-mini	multi	3	–	21.6	**−8.8**	–	**+27.9**	**+66.4**	**−8.4**	**+20.0**	–	−1.0
gpt-4o-mini	minimal	3	–	35.0	**−18.1**	–	**+38.4**	**+51.2**	–	–	–	–
gpt-4.1	single	1	20.0	64.7	**−8.0**	**−35.3**	**+26.7**	**+26.0**	–	−2.3	+7.3	−3.0
gpt-4.1	multi	1	–	13.0	**−6.3**	–	**+23.0**	**+78.3**	**−6.7**	**+17.0**	–	0.0
Qwen2.5-7B	single	1	4.3	50.3	−3.3	−5.0	**+25.3**	**+35.7**	−4.0	+7.3	–	−1.0
Qwen2.5-7B	multi	1	–	23.7	**−8.3**	–	**+17.7**	**+63.7**	−8.3	+6.7	–	−1.0
gpt-6-luna	single	3	16.9	53.6	**−9.4**	**−29.1**	**+36.4**	**+37.0**	**−9.3**	+8.6	−6.3	−1.3
gpt-6.1-terra	single	1	20.0	73.0	−2.3	−5.3	**+17.0**	**+18.3**	−2.7	−11.7	**−21.0**	0.0
Claude Haiku 5.5	single	3	11.8	66.8	+0.7	+6.6	**+23.3**	**+22.6**	+0.7	−4.1	+1.7	−5.0
Claude Sonnet 5.5	single	1	–	75.7	+3.0	+4.0	**+16.3**	–	–	−2.7	–	−1.0
gpt-4o-mini	forced	3	–	55.4	−0.2	**+7.6**	**+24.2**	–	–	+4.0	–	−2.7
gpt-6-luna	forced	3	–	64.6	+1.6	**+10.3**	**+26.4**	–	–	−2.4	–	−1.3
Claude Haiku 5.5	forced	3	–	69.3	**+3.9**	**+5.1**	**+21.1**	–	–	−4.7	–	−1.3
gpt-6.1-terra	forced	1	–	77.7	−0.3	**+6.7**	**+14.7**	–	–	**−14.3**	–	0.0
Majority misinformation: three wrong-side passages, one correct												
gpt-4o-mini	single	3	–	21.6	+1.1	−5.0	**+32.7**	**+65.1**	−2.2	**+5.7**	+6.8	−4.7
gpt-6-luna	single	3	–	29.8	**+6.6**	+0.3	**+60.4**	–	–	**+7.1**	–	−1.3
gpt-6.1-terra	single	1	–	26.0	−3.7	**+22.0**	**+66.3**	–	–	**+9.0**	–	0.0
Claude Haiku 5.5	single	3	–	31.0	+4.3	–	**+57.3**	–	–	**+6.8**	–	−5.0
RAGuard: real Reddit posts labelled supporting or misleading (294 conflict claims)												
gpt-4o-mini	single	3	64.9	36.7	+3.2	−5.7	**+9.9**	+5.9	–	+0.6	–	+1.7
gpt-6-luna	single	3	64.1	23.2	0.0	**−7.0**	**+8.8**	−1.0	–	+3.1	–	−1.3
Claude Haiku 5.5	single	3	65.1	24.5	+0.3	**+5.9**	**+12.7**	+2.0	–	**+5.0**	–	+1.3
gpt-6.1-terra	single	1	74.1	24.8	+2.4	+0.7	**+6.8**	−1.0	–	+2.0	–	+1.0
Claude Sonnet 5.5	single	1	–	27.2	+4.4	–	**+20.4**	–	–	**+7.1**	–	+10.0

Table 4: ConflictBank conflict questions (300; one correct answer each) and RAGuard conflict claims (294; answer true or false): accuracy (%, one-way strict) closed-book and with the passages but no flags, then the paired difference in points made by each format. *forced*: the one-answer prompt without the option to answer "unknown". *Prompt*: the answer policy: *single* says there is one answer, *multi* asks for every valid answer (our default skeleton), *minimal* states no policy. *One wrong*: the neutral flag plus "exactly one of the two contains misinformation". *Single call*: one call that names the suspect passages, then answers. Last column: the LLM directional flag's change on the 100 clean questions. The LLM detector is gpt-4o-mini for every generator. **Bold**: significant under the paper's single scheme.

this. Removal does not depend on the model acting on a warning, and it gains in both groups.

C.2 Real detectors

CAAG flags 34.1% of clean questions. Its neutral flags do not beat random flags at the same rate (H5; −0.4 vs. +1.0 points overall). The 90% CI of the difference, [−3.1, +0.1], rules out a gain larger than +0.1 points but does not establish equivalence within ±3 points (TOST). The directional-versus-neutral comparison with the LLM detector's pairs was registered as exploratory in the addendum. Under the bidirectional metric, the LLM detector's directional flag costs clean questions 6.1 points. With gpt-4.1 as the generator, its gain over all 300 questions (+5.4) is significant after Holm, but the 35 unlabelled questions contribute disproportionately to it (they gain 13.3 points); over the labelled questions it is +4.4 [+0.6, +8.3]. On ConflictBank the NLI cross-encoder flags no pair at all: ConflictBank passages are long, LLM-written and agree on everything but the one fact, and the truncated pair rarely contains it.

The single call with stronger models. Asked to name suspect passages before answering, the reasoning models and Claude Haiku distrust most of ConflictBank's generated passages and abstain, losing up to 51.0 points on clean questions (Table 34).

Reasoning models on RAMDocs. With no flags, gpt-6-luna and gpt-6.1-terra are strictly correct on few RAMDocs conflict questions because the default prompt asks for every valid answer and they list nearly every candidate: 2.8 and 3.0 answers on average, and 72% and 76% of their answers include the misinformation, while they recover 79% and 77% of the gold answers.

Detector parsing. The LLM detectors' JSON failed to parse on 1 of 2400 ConflictBank questions (a Claude Sonnet output), which then count

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	29.9	31.3	22.1	41.5	911
Prompt only					
Instruction (v1)	+1.6	+0.8	−1.2	+2.9	948
Instruction (v2)	+3.1	+2.3	+5.3	−1.1	951
Oracle detector					
Oracle, flag	−3.1	−3.5	**−6.7**	0.0	980
Oracle, directional flag	+6.2	+7.0	**+13.4**	0.0	998
Oracle, remove suspect	**+12.8**	**+14.5**	**+27.6**	0.0	825
Real detectors (exploratory)					
Random, flag	+1.0	+1.0	+0.7	+1.3	949
CAAG, flag	−0.4	−0.8	−0.7	−0.8	942
CAAG, remove	−3.3	−3.8	0.0	−7.9	809
NLI, flag	+0.2	+0.3	−0.5	+1.1	1021
NLI, remove	−5.7	−5.0	−2.2	−8.2	592
LLM, flag	−1.8	−1.3	−4.1	+1.9	970
LLM, directional flag	+5.0	+4.9	+12.5	−3.4	983
LLM, remove suspect	+0.3	0.0	+18.7	**−20.6**	718
Reference					
MADAM-RAG	+5.7	+5.7	+3.6	+7.9	8196

Table 5: RAMDocs hold-out, gpt-4o-mini, 3 seeds, one-way strict accuracy (primary). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

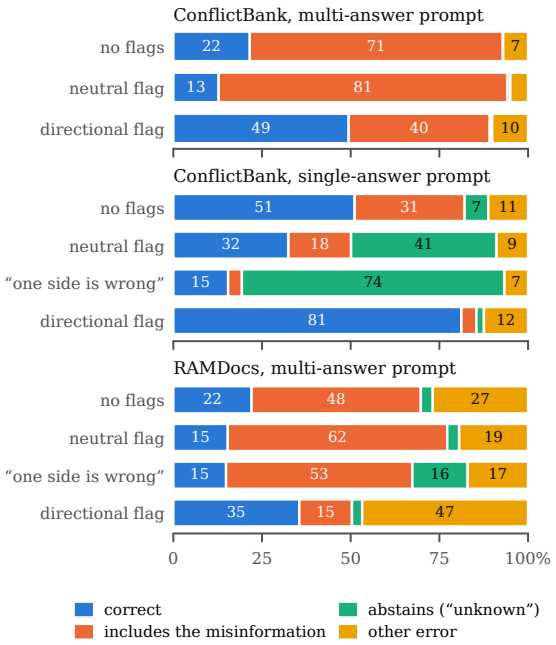


Figure 3: What a flag does to answers on conflict questions (gpt-4o-mini, oracle pairs; share of answers, %). The answer policy decides how a neutral flag fails: a model told to give every valid answer adds the misinformation, and one told there is one answer abstains. Only the directional flag raises the share of correct answers. The categories are exclusive and an answer the primary metric scores correct counts as correct, so the misinformation share here can be a little lower than in Table 1, which counts every answer containing a wrong answer under the bidirectional matcher.

as flagging nothing.

C.3 A detector from another family

Replacing the gpt-4o-mini detector with Qwen2.5-7B (gpt-4o-mini still generates) gives a much less selective detector. It flags 96.8% of clean RAMDocs questions, but its direction accuracy is 89.0%. Its directional flag gains +10.6 points on conflict questions and 0.0 on clean ones, and its neutral flag −2.6. Removal again trades a conflict-question gain (+14.1) for a clean-question loss (−14.6). The conclusions do not depend on the detector's family.

C.4 CONFLICTS

On the 237 automatically scorable CONFLICTS questions (exploratory, 3 seeds; most have no conflict), no condition moves accuracy significantly (Table 31).

D Reproducibility

Runs. Every number in this paper is produced by one script from committed run outputs: per-call responses, token counts, detector outputs and configurations. The pre-registration was committed before any confirmatory run. Its deviations are listed below.

Deviations from the pre-registration.

1. *Primary metric.* The first draft of the pre-registration named the bidirectional metric as primary. Before the confirmatory run we switched to one-way matching, because it reproduces MADAM-RAG's published score on the pilot split (Section 4); bidirectional became the robustness metric. Every table reports both.
2. *Second generator.* The plan named an open-weights model. None was available before the confirmatory run, so gpt-4.1 was used (H9). We later added Qwen2.5-7B-Instruct, registered before its run. Because local inference takes about 3 s per call, the Qwen runs use one seed; we made that change before inspecting any Qwen result.
3. *CONFLICTS judge.* The plan called for a judge

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H1: detection quality					
H1	F1 llm_dir > nli	+2.6 [−2.4, +7.5]	0.311 (0.415)	not supported	–
H1	F1 nli > caag	+12.2 [+7.5, +16.9]	<0.001 (<0.001)	supported	–
H1	direction accuracy llm_dir > 50% (cluster CI[r])	88.3% [83.3, 92.9]	<0.001 (<0.001)	supported	–
H2: a correct neutral flag					
H2	oracle+flag < plain (conflict)	−6.7 [−10.8, −3.1]	<0.001 (0.002)	supported	yes
H3: what direction adds					
H3	oracle+flag_directional > plain (conflict)	+13.4 [+6.7, +20.4]	<0.001 (<0.001)	supported	yes
H3	oracle+flag_directional > oracle+flag (conflict)	+20.1 [+13.9, +26.9]	<0.001 (<0.001)	supported	yes
H3	oracle+remove_suspect > plain (conflict)	+27.6 [+20.6, +35.0]	<0.001 (<0.001)	supported	yes
H4: false flags					
H4a	false-flag cells below plain on clean q., directional flag[r]	0 of 9 cells	Holm	not supported	yes
H4a	false-flag cells below plain on clean q., removal[r]	6 of 9 cells	Holm	supported	yes
H4b	harm grows with clean-q. exposure, Spearman over 18 cells[r]	$\rho = -0.39$ [−0.74, −0.12]	–	supported	yes
H4c	removal harms clean q. more than directional flag[r]	5 of 9 matched cells	Holm	count only	–
H5: the heuristic detector vs random					
H5	caag+flag vs random+flag (all)	−1.4 [−3.4, +0.3]	0.178 (0.258)	n.s.	yes
H5	equivalence within ±3 points (TOST)[r]	90% CI [−3.1, +0.1]	–	not equivalent	yes
H6: does the grid predict real detectors					
H6	actual effect inside the 95% interval of the document-level prediction[r]	7 of 8 (pair level: 6 of 8)	–	descriptive	–
H6	document-level error < pair-level error (paired, exact sign-flip)[r]	−1.2 [−3.6, +0.4]	0.414	not supported	yes
H7: prompt-only instructions					
H7	instruct vs plain (conflict)	−1.2 [−4.3, +1.7]	1.000 (0.636)	n.s.	yes
H7	instruct vs plain (clean)	+2.9 [0.0, +6.1]	0.280 (0.146)	n.s.	yes
H7	instruct_v2 vs plain (conflict)	+5.3 [+1.4, +9.6]	0.059 (0.028)	n.s.	yes
H7	instruct_v2 vs plain (clean)	−1.1 [−5.0, +2.6]	1.000 (0.787)	n.s.	yes
H9: gpt-4.1					
H9	gpt-4.1: oracle+flag < plain (conflict)	−8.6 [−13.2, −4.3]	<0.001 (<0.001)	supported	yes
H9	gpt-4.1: oracle+flag_directional > plain (conflict)	+18.2 [+12.0, +24.7]	<0.001 (<0.001)	supported	yes
H9	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	+26.9 [+20.1, +33.6]	<0.001 (<0.001)	supported	yes
H9	gpt-4.1: oracle+remove_suspect > plain (conflict)	+30.2 [+22.5, +37.9]	<0.001 (<0.001)	supported	yes
H8: genuine conflicts (WikiContradict)					
H8	WikiContradict: instruct vs plain	+8.7 [+5.4, +12.3]	<0.001 (<0.001)	significant	–
H8	WikiContradict: instruct_v2 vs plain	−6.7 [−10.0, −3.7]	<0.001 (<0.001)	significant	–
H8	WikiContradict: oracle+flag vs plain	+6.5 [+3.0, +10.0]	<0.001 (<0.001)	supported	–
H8	WikiContradict: caag+flag vs plain	+0.7 [0.0, +1.6]	0.250 (0.347)	n.s.	–
H8	WikiContradict: nli+flag vs plain	+3.2 [+0.3, +6.1]	0.077 (0.067)	n.s.	–
H8	WikiContradict: llm_dir+flag vs plain	+6.2 [+2.8, +9.6]	<0.001 (<0.001)	significant	–

Table 6: Pre-registered tests registered before the confirmatory run (RAMDocs hold-out unless stated; one-way strict accuracy; differences in points). Rows marked [r] are re-analyses registered later, before they were computed. Directional predictions are *supported* or *not supported*; tests with no predicted sign are *significant* or *n.s.* *Robust*: the same outcome under the bidirectional metric (robustness to the RAMDocs label rule is in Table 29, where H4b's correlation weakens). p-values come from 10,000 sign-flip permutations, extended to 200,000 when none reaches the observed statistic, Holm-adjusted within the family; this is the significance used everywhere in the paper. In brackets: the Benjamini–Hochberg q over all registered tests of the programme (primary metric).

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H10: without the answer policy (minimal skeleton)					
H10	gpt-4o-mini: oracle+flag vs plain (conflict)	-5.3 [-9.4, -1.4]	0.042 (0.020)	significant	yes
H10	gpt-4o-mini: oracle+flag_directional > plain (conflict)	$+5.0$ [-1.7, $+11.8$]	0.314 (0.232)	not supported	yes
H10	gpt-4o-mini: oracle+flag_directional > oracle+flag (conflict)	$+10.3$ [$+4.8$, $+16.1$]	0.005 (0.001)	supported	yes
H10	gpt-4o-mini: oracle+remove_suspect > plain (conflict)	$+18.9$ [$+12.5$, $+25.9$]	<0.001 (<0.001)	supported	yes
H10	gpt-4.1: oracle+flag vs plain (conflict)	-11.0 [-17.5, -4.6]	0.010 (0.003)	significant	yes
H10	gpt-4.1: oracle+flag_directional > plain (conflict)	$+12.7$ [$+5.8$, $+19.7$]	0.005 (0.001)	supported	yes
H10	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	$+23.7$ [$+17.0$, $+30.7$]	<0.001 (<0.001)	supported	yes
H10	gpt-4.1: oracle+remove_suspect > plain (conflict)	$+14.9$ [$+8.6$, $+21.6$]	<0.001 (<0.001)	supported	yes
H10	Qwen2.5-7B: oracle+flag vs plain (conflict)	-5.8 [-10.8, -1.4]	0.117 (0.067)	n.s.	yes
H10	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	$+2.9$ [-3.6, $+9.4$]	0.509 (0.614)	not supported	yes
H10	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	$+8.6$ [$+2.9$, $+14.4$]	0.041 (0.016)	supported	yes
H10	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	$+10.8$ [$+4.3$, $+17.3$]	0.017 (0.006)	supported	yes
H11: an open-weights generator					
H11	Qwen2.5-7B: oracle+flag < plain (conflict)	-5.8 [-10.8, -0.7]	0.114 (0.092)	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	$+3.6$ [-0.7, $+8.6$]	0.224 (0.320)	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	$+9.4$ [$+5.0$, $+14.4$]	0.001 (0.001)	supported	yes
H11	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	$+18.0$ [$+10.8$, $+25.2$]	<0.001 (<0.001)	supported	yes

Table 7: Pre-registered tests added in the first addendum (§9; H10–H11), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H12: ConflictBank					
H12	gpt-4o-mini: neutral flag vs plain (conflict)	−8.8 [−12.8, −4.9]	<0.001 (<0.001)	significant	yes
H12	gpt-4o-mini: dir. flag > plain (conflict)	+27.9 [+22.8, +32.9]	<0.001 (<0.001)	supported	yes
H12	gpt-4o-mini: dir. flag > neutral flag (conflict)	+36.7 [+31.6, +41.9]	<0.001 (<0.001)	supported	yes
H12	gpt-4o-mini: removal > plain (conflict)	+66.4 [+61.1, +71.7]	<0.001 (<0.001)	supported	yes
H12	gpt-4.1: neutral flag vs plain (conflict)	−6.3 [−9.7, −3.3]	<0.001 (<0.001)	significant	yes
H12	gpt-4.1: dir. flag > plain (conflict)	+23.0 [+18.0, +28.3]	<0.001 (<0.001)	supported	yes
H12	gpt-4.1: dir. flag > neutral flag (conflict)	+29.3 [+24.0, +34.7]	<0.001 (<0.001)	supported	yes
H12	gpt-4.1: removal > plain (conflict)	+78.3 [+73.3, +83.0]	<0.001 (<0.001)	supported	yes
H12	Qwen2.5-7B: neutral flag vs plain (conflict)	−8.3 [−12.3, −4.3]	<0.001 (<0.001)	significant	yes
H12	Qwen2.5-7B: dir. flag > plain (conflict)	+17.7 [+12.0, +23.3]	<0.001 (<0.001)	supported	yes
H12	Qwen2.5-7B: dir. flag > neutral flag (conflict)	+26.0 [+20.7, +31.3]	<0.001 (<0.001)	supported	yes
H12	Qwen2.5-7B: removal > plain (conflict)	+63.7 [+58.3, +69.0]	<0.001 (<0.001)	supported	yes
H13: neutral flag + resolution instruction					
H13	RAMDocs, gpt-4o-mini: neutral flag + instruction vs plain (conflict)	−2.4 [−6.2, +1.2]	0.743 (0.354)	n.s.	yes
H13	RAMDocs, gpt-4o-mini: dir. flag + instruction > neutral flag + instruction (conflict)	+15.6 [+9.6, +21.8]	<0.001 (<0.001)	supported	yes
H13	RAMDocs, gpt-4.1: neutral flag + instruction vs plain (conflict)	+5.8 [−2.9, +14.4]	0.743 (0.346)	n.s.	yes
H13	RAMDocs, gpt-4.1: dir. flag + instruction > neutral flag + instruction (conflict)	+25.2 [+18.0, +32.4]	<0.001 (<0.001)	supported	yes
H13	ConflictBank, gpt-4o-mini: neutral flag + instruction vs plain (conflict)	+1.6 [−2.8, +6.0]	0.743 (0.622)	n.s.	yes
H13	ConflictBank, gpt-4o-mini: dir. flag + instruction > neutral flag + instruction (conflict)	+62.2 [+57.1, +67.2]	<0.001 (<0.001)	supported	yes
H14: published baselines					
H14	RAMDocs, gpt-4o-mini: LLM dir. flag vs astute multi (labelled)	+10.8 [+5.9, +16.0]	<0.001 (<0.001)	significant	yes
H14	RAMDocs, gpt-4o-mini: LLM dir. flag vs trustrag multi (labelled)	+23.0 [+18.1, +28.3]	<0.001 (<0.001)	significant	yes
H14	RAMDocs, gpt-4.1: LLM dir. flag vs astute multi (labelled)	+6.4 [+0.4, +12.5]	0.151 (0.083)	n.s.	no
H14	RAMDocs, gpt-4.1: LLM dir. flag vs trustrag multi (labelled)	+19.6 [+13.6, +25.7]	<0.001 (<0.001)	significant	yes
H14	ConflictBank, gpt-4o-mini: LLM dir. flag vs astute (all questions)	−3.4 [−8.1, +1.3]	0.324 (0.238)	n.s.	yes
H14	ConflictBank, gpt-4o-mini: LLM dir. flag vs trustrag (all questions)	+0.8 [−3.9, +5.6]	0.761 (0.843)	n.s.	yes

Table 8: Pre-registered tests added in the second (§10; H12–H14), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H15: ConflictBank, single-answer prompt					
H15	ConflictBank, gpt-4o-mini: neutral flag vs plain (conflict)	−18.7 [−23.4, −14.0]	<0.001 (<0.001)	significant	yes
H15	ConflictBank, gpt-4o-mini: dir. flag > neutral flag (conflict)	+48.8 [+43.6, +54.1]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4o-mini: dir. flag > plain (conflict)	+30.1 [+25.1, +35.2]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4o-mini: LLM dir. flag > plain (all q.)	+5.2 [+1.0, +9.3]	0.065 (0.030)	not supported	no
H15	ConflictBank, gpt-4.1: neutral flag vs plain (conflict)	−8.0 [−12.7, −3.7]	0.006 (0.003)	significant	yes
H15	ConflictBank, gpt-4.1: dir. flag > neutral flag (conflict)	+34.7 [+29.0, +40.3]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4.1: dir. flag > plain (conflict)	+26.7 [+21.7, +31.7]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, gpt-4.1: LLM dir. flag > plain (all q.)	−2.5 [−6.5, +1.5]	0.522 (0.359)	not supported	yes
H15	ConflictBank, Qwen2.5-7B: neutral flag vs plain (conflict)	−3.3 [−9.0, +2.0]	0.522 (0.393)	n.s.	yes
H15	ConflictBank, Qwen2.5-7B: dir. flag > neutral flag (conflict)	+28.7 [+23.3, +34.0]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, Qwen2.5-7B: dir. flag > plain (conflict)	+25.3 [+19.7, +31.0]	<0.001 (<0.001)	supported	yes
H15	ConflictBank, Qwen2.5-7B: LLM dir. flag > plain (all q.)	+5.2 [+0.5, +10.0]	0.130 (0.073)	not supported	yes
H16: information vs direction (one-wrong flag)					
H16	ConflictBank, gpt-4o-mini: dir. flag > one-wrong flag (conflict)	+65.8 [+60.6, +70.9]	<0.001 (<0.001)	supported	yes
H16	ConflictBank, gpt-4o-mini: one-wrong flag vs plain (conflict)	−35.7 [−41.1, −30.3]	<0.001 (<0.001)	significant	yes
H16	ConflictBank, gpt-4.1: dir. flag > one-wrong flag (conflict)	+62.0 [+56.0, +67.7]	<0.001 (<0.001)	supported	yes
H16	ConflictBank, gpt-4.1: one-wrong flag vs plain (conflict)	−35.3 [−41.3, −29.3]	<0.001 (<0.001)	significant	yes
H16	RAMDocs, gpt-4o-mini: dir. flag > one-wrong flag (conflict)	+20.6 [+14.4, +27.3]	<0.001 (<0.001)	supported	yes
H16	RAMDocs, gpt-4o-mini: one-wrong flag vs plain (conflict)	−7.2 [−12.2, −2.6]	0.006 (0.011)	significant	yes
H17: majority misinformation					
H17	CB majority, gpt-4o-mini: neutral flag vs plain (conflict)	+1.1 [−2.7, +4.9]	0.603 (0.702)	n.s.	yes
H17	CB majority, gpt-4o-mini: dir. flag > plain (conflict)	+32.7 [+27.3, +37.9]	<0.001 (<0.001)	supported	yes
H17	CB majority, gpt-4o-mini: LLM dir. flag vs plain (conflict)	+5.7 [+1.4, +9.9]	0.017 (0.016)	significant	yes
H18: a single-call baseline					
H18	RAMDocs, gpt-4o-mini: LLM dir. flag vs self-identify (labelled)	−1.3 [−4.8, +2.1]	1.000 (0.627)	n.s.	yes
H18	ConflictBank, gpt-4o-mini: LLM dir. flag vs self-identify (all questions)	−0.2 [−4.2, +3.7]	1.000 (0.995)	n.s.	yes
H18	CB majority, gpt-4o-mini: LLM dir. flag vs self-identify (all questions)	−1.8 [−4.6, +1.1]	0.730 (0.341)	n.s.	yes
H19: current reasoning models					
H19	ConflictBank, gpt-6-luna: neutral flag vs plain (conflict)	−9.4 [−12.9, −6.0]	<0.001 (<0.001)	significant	yes
H19	ConflictBank, gpt-6-luna: dir. flag > neutral flag (conflict)	+45.9 [+40.8, +50.9]	<0.001 (<0.001)	supported	yes

Table 9: Pre-registered tests added in the third and fourth (§11–§12; H15–H19) (part 1 of 2), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H19: current reasoning models					
H19	ConflictBank, gpt-6-luna: dir. flag > one-wrong flag (conflict)	+65.6 [+60.7, +70.1]	<0.001 (<0.001)	supported	yes
H19	ConflictBank, gpt-6-luna: LLM dir. flag > plain (all q.)	+6.1 [+2.1, +10.0]	0.019 (0.008)	supported	yes
H19	ConflictBank, gpt-6.1-terra: neutral flag vs plain (conflict)	−2.3 [−6.7, +2.0]	0.717 (0.471)	n.s.	yes
H19	ConflictBank, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+19.3 [+14.7, +24.0]	<0.001 (<0.001)	supported	yes
H19	ConflictBank, gpt-6.1-terra: dir. flag > one-wrong flag (conflict)	+22.3 [+17.7, +27.3]	<0.001 (<0.001)	supported	yes
H19	ConflictBank, gpt-6.1-terra: LLM dir. flag > plain (all q.)	−8.8 [−13.2, −4.2]	0.002 (<0.001)	not supported	yes
H19	RAMDocs, gpt-6-luna: neutral flag vs plain (conflict)	−3.1 [−5.3, −1.4]	0.007 (0.003)	significant	yes
H19	RAMDocs, gpt-6-luna: dir. flag > neutral flag (conflict)	+42.2 [+34.5, +49.9]	<0.001 (<0.001)	supported	yes
H19	RAMDocs, gpt-6.1-terra: neutral flag vs plain (conflict)	−0.7 [−2.9, +1.4]	1.000 (1.000)	n.s.	yes
H19	RAMDocs, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+40.3 [+32.4, +48.2]	<0.001 (<0.001)	supported	yes
H19	CB majority, gpt-6-luna: dir. flag > plain (conflict)	+60.4 [+55.3, +65.4]	<0.001 (<0.001)	supported	yes
H19	CB majority, gpt-6-luna: LLM dir. flag vs plain (conflict)	+7.1 [+2.4, +11.8]	0.019 (0.008)	significant	yes
H19	CB majority, gpt-6.1-terra: dir. flag > plain (conflict)	+66.3 [+61.0, +71.7]	<0.001 (<0.001)	supported	yes
H19	CB majority, gpt-6.1-terra: LLM dir. flag vs plain (conflict)	+9.0 [+3.3, +14.7]	0.019 (0.008)	significant	yes

Table 10: Pre-registered tests added in the third and fourth (§11–§12; H15–H19) (part 2 of 2), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H20: forced choice (no "unknown")					
H20	ConflictBank forced, gpt-4o-mini: neutral flag vs plain (conflict)	-0.2 [-4.1, $+3.7$]	1.000 (1.000)	n.s.	yes
H20	ConflictBank forced, gpt-4o-mini: one-wrong flag vs plain (conflict)	$+7.6$ [$+2.7$, $+12.3$]	0.024 (0.007)	significant	yes
H20	ConflictBank forced, gpt-4o-mini: dir. flag > plain (conflict)	$+24.2$ [$+19.7$, $+29.0$]	<0.001 (<0.001)	supported	yes
H20	ConflictBank forced, gpt-6-luna: neutral flag vs plain (conflict)	$+1.6$ [-1.6, $+4.8$]	1.000 (0.479)	n.s.	yes
H20	ConflictBank forced, gpt-6-luna: one-wrong flag vs plain (conflict)	$+10.3$ [$+5.6$, $+15.1$]	<0.001 (<0.001)	significant	yes
H20	ConflictBank forced, gpt-6-luna: dir. flag > plain (conflict)	$+26.4$ [$+21.9$, $+31.0$]	<0.001 (<0.001)	supported	yes
H20	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain (conflict)	-0.3 [-4.0, $+3.3$]	1.000 (1.000)	n.s.	yes
H20	ConflictBank forced, gpt-6.1-terra: one-wrong flag vs plain (conflict)	$+6.7$ [$+2.3$, $+11.3$]	0.034 (0.013)	significant	yes
H20	ConflictBank forced, gpt-6.1-terra: dir. flag > plain (conflict)	$+14.7$ [$+10.3$, $+19.0$]	<0.001 (<0.001)	supported	yes
H20	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain (conflict)	$+3.9$ [$+1.2$, $+6.7$]	0.034 (0.011)	significant	yes
H20	ConflictBank forced, claude-haiku-5-5: one-wrong flag vs plain (conflict)	$+5.1$ [$+1.4$, $+8.8$]	0.034 (0.016)	significant	yes
H20	ConflictBank forced, claude-haiku-5-5: dir. flag > plain (conflict)	$+21.1$ [$+16.9$, $+25.4$]	<0.001 (<0.001)	supported	yes
H22: pair topology					
H22	RAMDocs, gpt-4o-mini: one-wrong without pairs vs with pairs (conflict)	$+6.5$ [$+2.4$, $+10.8$]	0.015 (0.005)	significant	yes
H22	RAMDocs, gpt-6-luna: one-wrong without pairs vs with pairs (conflict)	$+10.1$ [$+5.8$, $+14.9$]	<0.001 (<0.001)	significant	yes
H22	RAMDocs, gpt-6.1-terra: one-wrong without pairs vs with pairs (conflict)	-0.7 [-6.5, $+5.8$]	1.000 (1.000)	n.s.	yes
H22	CB majority, gpt-4o-mini: neutral without pairs vs with pairs (conflict)	-3.1 [-6.4, $+0.2$]	0.390 (0.124)	n.s.	yes
H22	CB majority, gpt-4o-mini: LLM dir. flag without pairs vs with pairs (conflict)	$+0.3$ [-2.1, $+2.8$]	1.000 (0.935)	n.s.	yes
H22	CB majority, gpt-6-luna: neutral without pairs vs with pairs (conflict)	-6.1 [-9.1, -3.1]	0.002 (<0.001)	significant	yes
H22	CB majority, gpt-6-luna: LLM dir. flag without pairs vs with pairs (conflict)	$+0.8$ [-1.7, $+3.1$]	1.000 (0.683)	n.s.	yes
H22	CB majority, gpt-6.1-terra: neutral without pairs vs with pairs (conflict)	$+10.0$ [$+5.7$, $+14.3$]	0.002 (<0.001)	significant	yes
H22	CB majority, gpt-6.1-terra: LLM dir. flag without pairs vs with pairs (conflict)	$+1.7$ [-1.3, $+4.7$]	1.000 (0.475)	n.s.	yes
H23: answer key					
H23	RAMDocs, gpt-4o-mini: dir. flag vs answer key (conflict)	-1.7 [-6.2, $+3.1$]	1.000 (0.656)	n.s.	yes
H23	RAMDocs, gpt-6-luna: dir. flag vs answer key (conflict)	-4.1 [-9.1, $+0.7$]	0.808 (0.203)	n.s.	yes
H23	RAMDocs, gpt-6.1-terra: dir. flag vs answer key (conflict)	-5.0 [-10.8, $+0.7$]	0.827 (0.242)	n.s.	yes
H23	ConflictBank, gpt-4o-mini: dir. flag vs answer key (conflict)	$+5.3$ [$+1.6$, $+9.1$]	0.059 (0.013)	n.s.	no
H23	ConflictBank, gpt-6-luna: dir. flag vs answer key (conflict)	$+0.1$ [-1.6, $+1.8$]	1.000 (1.000)	n.s.	yes

Table 11: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 1 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H23: answer key					
H23	ConflictBank, gpt-6.1-terra: dir. flag vs answer key (conflict)	-1.7 [-3.3, -0.3]	0.515 (0.104)	n.s.	yes
H23	CB majority, gpt-4o-mini: dir. flag vs answer key (conflict)	$+3.2$ [-1.8, $+8.2$]	0.905 (0.320)	n.s.	yes
H23	CB majority, gpt-6-luna: dir. flag vs answer key (conflict)	$+0.4$ [-1.2, $+2.2$]	1.000 (0.798)	n.s.	yes
H23	CB majority, gpt-6.1-terra: dir. flag vs answer key (conflict)	$+2.3$ [$+0.3$, $+4.7$]	0.515 (0.115)	n.s.	yes
H25: Claude					
H25	ConflictBank, claude-haiku-5-5: neutral flag vs plain (conflict)	$+0.7$ [-2.2, $+3.4$]	1.000 (0.790)	n.s.	yes
H25	ConflictBank, claude-haiku-5-5: dir. flag > neutral flag (conflict)	$+22.7$ [$+18.6$, $+26.9$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-haiku-5-5: dir. flag > one-wrong flag (conflict)	$+16.8$ [$+13.0$, $+20.8$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-haiku-5-5: LLM dir. flag > plain (all q.)	-4.3 [-8.1, -0.7]	0.145 (0.043)	not supported	yes
H25	ConflictBank, claude-sonnet-5-5: neutral flag vs plain (conflict)	$+3.0$ [-0.3, $+6.3$]	0.634 (0.193)	n.s.	yes
H25	ConflictBank, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	$+13.3$ [$+9.7$, $+17.3$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-sonnet-5-5: dir. flag > one-wrong flag (conflict)	$+12.3$ [$+8.7$, $+16.3$]	<0.001 (<0.001)	supported	yes
H25	ConflictBank, claude-sonnet-5-5: LLM dir. flag > plain (all q.)	-2.2 [-5.8, $+1.0$]	0.993 (0.346)	not supported	yes
H25	RAMDocs, claude-haiku-5-5: neutral flag vs plain (conflict)	$+1.2$ [-2.6, $+5.3$]	1.000 (0.732)	n.s.	yes
H25	RAMDocs, claude-haiku-5-5: dir. flag > neutral flag (conflict)	$+27.1$ [$+20.1$, $+34.1$]	<0.001 (<0.001)	supported	yes
H25	RAMDocs, claude-sonnet-5-5: neutral flag vs plain (conflict)	0.0 [-4.3, $+4.3$]	1.000 (1.000)	n.s.	yes
H25	RAMDocs, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	$+14.4$ [$+7.9$, $+21.6$]	0.002 (<0.001)	supported	yes
H25	CB majority, claude-haiku-5-5: dir. flag > plain (conflict)	$+57.3$ [$+52.0$, $+62.7$]	<0.001 (<0.001)	supported	yes
H25	CB majority, claude-haiku-5-5: LLM dir. flag vs plain (conflict)	$+6.8$ [$+2.1$, $+11.4$]	0.043 (0.012)	significant	yes
H26: RAGuard (real misleading posts)					
H26	RAGuard, gpt-4o-mini: neutral flag vs plain (conflict)	$+3.2$ [$+0.1$, $+6.3$]	0.426 (0.087)	n.s.	yes
H26	RAGuard, gpt-4o-mini: dir. flag > plain (conflict)	$+9.9$ [$+5.6$, $+14.2$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, gpt-4o-mini: dir. flag > neutral flag (conflict)	$+6.7$ [$+2.9$, $+10.5$]	0.008 (0.001)	supported	yes
H26	RAGuard, gpt-4o-mini: one-wrong flag vs plain (conflict)	-5.7 [-10.2, -1.2]	0.161 (0.030)	n.s.	yes
H26	RAGuard, gpt-4o-mini: LLM dir. flag vs plain (conflict)	$+0.6$ [-2.8, $+4.0$]	1.000 (0.878)	n.s.	yes
H26	RAGuard, gpt-6-luna: neutral flag vs plain (conflict)	0.0 [-1.7, $+1.7$]	1.000 (1.000)	n.s.	yes
H26	RAGuard, gpt-6-luna: dir. flag > plain (conflict)	$+8.8$ [$+5.7$, $+12.1$]	<0.001 (<0.001)	supported	yes
H26	RAGuard, gpt-6-luna: dir. flag > neutral flag (conflict)	$+8.8$ [$+5.7$, $+12.2$]	<0.001 (<0.001)	supported	yes

Table 12: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 2 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H26: RAGuard (real misleading posts)					
H26	RAGuard, gpt-6-luna: one-wrong flag vs plain (conflict)	−7.0 [−10.3, −3.9]	<0.001 (<0.001)	significant	yes
H26	RAGuard, gpt-6-luna: LLM dir. flag vs plain (conflict)	+3.1 [−0.1, +6.2]	0.511 (0.117)	n.s.	yes
H26	RAGuard, gpt-6.1-terra: neutral flag vs plain (conflict)	+2.4 [0.0, +5.1]	0.731 (0.189)	n.s.	yes
H26	RAGuard, gpt-6.1-terra: dir. flag > plain (conflict)	+6.8 [+3.4, +10.5]	0.005 (<0.001)	supported	yes
H26	RAGuard, gpt-6.1-terra: dir. flag > neutral flag (conflict)	+4.4 [+1.4, +7.8]	0.126 (0.022)	not supported	yes
H26	RAGuard, gpt-6.1-terra: one-wrong flag vs plain (conflict)	+0.7 [−3.4, +4.8]	1.000 (0.935)	n.s.	yes
H26	RAGuard, gpt-6.1-terra: LLM dir. flag vs plain (conflict)	+2.0 [−1.0, +5.1]	1.000 (0.393)	n.s.	yes
H26	RAGuard, claude-haiku-5-5: neutral flag vs plain (conflict)	+0.3 [−2.0, +2.7]	1.000 (0.933)	n.s.	yes
H26	RAGuard, claude-haiku-5-5: dir. flag > plain (conflict)	+12.7 [+9.2, +16.3]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-haiku-5-5: dir. flag > neutral flag (conflict)	+12.4 [+9.0, +15.9]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-haiku-5-5: one-wrong flag vs plain (conflict)	+5.9 [+2.9, +9.0]	0.005 (<0.001)	significant	yes
H26	RAGuard, claude-haiku-5-5: LLM dir. flag vs plain (conflict)	+5.0 [+1.8, +8.2]	0.025 (0.005)	significant	yes
H26	RAGuard, claude-sonnet-5-5: neutral flag vs plain (conflict)	+4.4 [+0.7, +8.2]	0.321 (0.062)	n.s.	yes
H26	RAGuard, claude-sonnet-5-5: dir. flag > plain (conflict)	+20.4 [+15.6, +25.2]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-sonnet-5-5: dir. flag > neutral flag (conflict)	+16.0 [+11.6, +20.4]	<0.001 (<0.001)	supported	yes
H26	RAGuard, claude-sonnet-5-5: LLM dir. flag vs plain (conflict)	+7.1 [+3.1, +11.2]	0.022 (0.004)	significant	yes
H21: risk and coverage					
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, misinformation rate (conflict)	−13.3 [−17.6, −9.1]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, misinformation rate (conflict)	−26.8 [−31.6, −22.0]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, misinformation rate (conflict)	−6.9 [−9.9, −4.0]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, misinformation rate (conflict)	−24.3 [−28.6, −20.2]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, misinformation rate (conflict)	−2.3 [−6.0, +1.3]	1.000 (0.398)	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, misinformation rate (conflict)	−17.3 [−21.7, −13.0]	<0.001 (<0.001)	significant	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, misinformation rate (conflict)	+0.1 [−2.6, +2.8]	1.000 (1.000)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, misinformation rate (conflict)	−21.8 [−26.1, −17.6]	<0.001 (<0.001)	significant	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, misinformation rate (conflict)	+0.3 [−3.4, +4.2]	1.000 (0.975)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, misinformation rate (conflict)	−24.9 [−29.7, −20.1]	<0.001 (<0.001)	significant	–

Table 13: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 3 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H21: risk and coverage					
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, misinformation rate (conflict)	−3.8 [−6.8, −0.9]	0.079 (0.030)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, misinformation rate (conflict)	−25.7 [−30.0, −21.3]	<0.001 (<0.001)	significant	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, misinformation rate (conflict)	−1.3 [−5.0, +2.3]	1.000 (0.695)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, misinformation rate (conflict)	−14.3 [−18.3, −10.3]	<0.001 (<0.001)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, misinformation rate (conflict)	−4.1 [−6.9, −1.3]	0.025 (0.008)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, misinformation rate (conflict)	−20.9 [−25.1, −16.8]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, abstention rate (conflict)	+34.3 [+29.3, +39.2]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, abstention rate (conflict)	−4.6 [−7.2, −2.0]	0.013 (0.002)	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, abstention rate (conflict)	+17.0 [+13.6, +20.6]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, abstention rate (conflict)	−11.1 [−14.1, −8.2]	<0.001 (<0.001)	significant	–
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, abstention rate (conflict)	+3.0 [+0.7, +5.3]	0.281 (0.042)	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, abstention rate (conflict)	−1.3 [−2.7, −0.3]	1.000 (0.189)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, abstention rate (conflict)	−0.3 [−0.9, +0.1]	1.000 (0.479)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, abstention rate (conflict)	−0.1 [−0.9, +0.8]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, abstention rate (conflict)	+0.3 [−0.1, +0.9]	1.000 (0.479)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, abstention rate (conflict)	−0.1 [−0.3, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, abstention rate (conflict)	0.0 [0.0, 0.0]	1.000 (1.000)	n.s.	–
H21	ConflictBank, gpt-4o-mini: neutral flag vs plain, selective accuracy (ratio) (conflict)	+0.2 [−4.8, +5.2]	1.000 (0.995)	n.s.	–
H21	ConflictBank, gpt-4o-mini: dir. flag vs plain, selective accuracy (ratio) (conflict)	+28.2 [+22.8, +33.5]	0.002 (<0.001)	significant	–
H21	ConflictBank, gpt-6-luna: neutral flag vs plain, selective accuracy (ratio) (conflict)	+1.2 [−2.2, +4.8]	1.000 (0.599)	n.s.	–
H21	ConflictBank, gpt-6-luna: dir. flag vs plain, selective accuracy (ratio) (conflict)	+29.9 [+25.0, +34.7]	0.002 (<0.001)	significant	–

Table 14: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 4 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H21: risk and coverage					
H21	ConflictBank, gpt-6.1-terra: neutral flag vs plain, selective accuracy (ratio) (conflict)	-0.1 [-4.0, $+3.7$]	1.000 (1.000)	n.s.	–
H21	ConflictBank, gpt-6.1-terra: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+16.1$ [$+11.5$, $+20.6$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank, claude-haiku-5-5: neutral flag vs plain, selective accuracy (ratio) (conflict)	$+0.4$ [-2.3, $+3.3$]	1.000 (0.860)	n.s.	–
H21	ConflictBank, claude-haiku-5-5: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+23.5$ [$+19.2$, $+27.9$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, gpt-4o-mini: neutral flag vs plain, selective accuracy (ratio) (conflict)	0.0 [-3.9, $+3.9$]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-4o-mini: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+24.2$ [$+19.4$, $+28.9$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, gpt-6-luna: neutral flag vs plain, selective accuracy (ratio) (conflict)	$+1.6$ [-1.7, $+4.7$]	1.000 (0.479)	n.s.	–
H21	ConflictBank forced, gpt-6-luna: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+26.4$ [$+21.9$, $+31.1$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, gpt-6.1-terra: neutral flag vs plain, selective accuracy (ratio) (conflict)	-0.3 [-4.0, $+3.3$]	1.000 (1.000)	n.s.	–
H21	ConflictBank forced, gpt-6.1-terra: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+14.7$ [$+10.3$, $+19.0$]	0.002 ($<$0.001)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: neutral flag vs plain, selective accuracy (ratio) (conflict)	$+3.9$ [$+1.2$, $+6.7$]	0.046 (0.011)	significant	–
H21	ConflictBank forced, claude-haiku-5-5: dir. flag vs plain, selective accuracy (ratio) (conflict)	$+21.1$ [$+17.0$, $+25.4$]	0.002 ($<$0.001)	significant	–
H24: detector versus generator reliability (crossed)					
H24	gain rises with detector-minus-generator reliability, Spearman over 30 cells	$\rho = +0.64$	$<$0.001	supported	–
H24	per-question slope of gain on reliability gap, within cells	$+47.6$ [$+43.2$, $+51.9$]	$<$0.001	supported	–

Table 15: Pre-registered tests added in the fifth and sixth (§13–§14; H20–H26) (part 5 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	Qwen2.5-7B, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+8.7 [+2.7, +14.3]	0.093 (0.011)	n.s.	no
H27	Qwen2.5-7B, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+4.3 [−1.7, +10.3]	1.000 (0.288)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+4.3 [−1.3, +10.0]	1.000 (0.252)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	−4.0 [−9.7, +1.7]	1.000 (0.288)	n.s.	yes
H27	claude-haiku-5-5, detector claude-haiku-5-5: LLM dir. flag vs hint (conflict)	+11.1 [+7.2, +15.1]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector claude-sonnet-5-5: LLM dir. flag vs hint (conflict)	+14.6 [+9.6, +19.7]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+14.1 [+9.8, +18.6]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	−1.4 [−6.0, +3.2]	1.000 (0.682)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+10.3 [+6.6, +14.1]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+11.0 [+7.1, +15.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+13.0 [+8.0, +18.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+0.7 [−4.7, +6.0]	1.000 (0.967)	n.s.	yes
H27	gpt-4.1, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+9.7 [+4.7, +14.7]	0.006 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+1.7 [−3.0, +6.3]	1.000 (0.688)	n.s.	yes
H27	gpt-4o-mini, detector claude-haiku-5-5: LLM dir. flag vs hint (conflict)	+6.2 [+1.1, +11.4]	0.279 (0.038)	n.s.	yes
H27	gpt-4o-mini, detector claude-sonnet-5-5: LLM dir. flag vs hint (conflict)	+5.8 [+0.2, +11.3]	0.485 (0.076)	n.s.	yes
H27	gpt-4o-mini, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+11.8 [+6.6, +17.0]	0.005 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+7.1 [+1.8, +12.6]	0.180 (0.022)	n.s.	yes
H27	gpt-4o-mini, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+8.1 [+2.9, +13.2]	0.037 (0.005)	significant	yes
H27	gpt-4o-mini, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+0.1 [−4.9, +5.2]	1.000 (1.000)	n.s.	yes
H27	gpt-6-luna, detector claude-haiku-5-5: LLM dir. flag vs hint (conflict)	+4.9 [−0.4, +10.3]	0.785 (0.138)	n.s.	yes
H27	gpt-6-luna, detector claude-sonnet-5-5: LLM dir. flag vs hint (conflict)	+3.9 [−0.8, +8.6]	0.992 (0.189)	n.s.	yes
H27	gpt-6-luna, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+13.7 [+8.4, +18.9]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	+6.0 [+0.6, +11.7]	0.485 (0.070)	n.s.	yes
H27	gpt-6-luna, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+8.2 [+3.7, +12.8]	0.005 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	−5.8 [−10.6, −1.0]	0.279 (0.036)	n.s.	yes

Table 16: Pre-registered tests added in the seventh (§15; H27–H29) (part 1 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	gpt-6.1-terra, detector gpt-4.1: LLM dir. flag vs hint (conflict)	+10.7 [+5.0, +16.0]	0.008 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4o-mini: LLM dir. flag vs hint (conflict)	−6.3 [−12.0, −0.7]	0.485 (0.071)	n.s.	yes
H27	gpt-6.1-terra, detector gpt-6-luna: LLM dir. flag vs hint (conflict)	+9.7 [+4.7, +14.7]	0.005 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-6.1-terra: LLM dir. flag vs hint (conflict)	+9.7 [+5.0, +14.3]	0.008 (<0.001)	significant	yes
H27	Qwen2.5-7B, detector gpt-4.1: hint vs plain (conflict)	+7.3 [+2.3, +12.3]	0.071 (0.011)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-4o-mini: hint vs plain (conflict)	+3.0 [−2.0, +8.0]	1.000 (0.402)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6-luna: hint vs plain (conflict)	+7.7 [+3.0, +12.7]	0.038 (0.006)	significant	no
H27	Qwen2.5-7B, detector gpt-6.1-terra: hint vs plain (conflict)	+12.0 [+7.0, +17.0]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector claude-haiku-5-5: hint vs plain (conflict)	−5.4 [−8.8, −2.1]	0.032 (0.004)	significant	yes
H27	claude-haiku-5-5, detector claude-sonnet-5-5: hint vs plain (conflict)	−6.3 [−9.4, −3.2]	0.004 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4.1: hint vs plain (conflict)	−3.1 [−5.8, −0.4]	0.341 (0.054)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-4o-mini: hint vs plain (conflict)	−2.7 [−5.6, +0.1]	0.681 (0.121)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-6-luna: hint vs plain (conflict)	−5.3 [−8.2, −2.6]	0.011 (0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6.1-terra: hint vs plain (conflict)	−6.9 [−9.9, −4.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4.1: hint vs plain (conflict)	−1.3 [−3.3, +0.3]	1.000 (0.393)	n.s.	yes
H27	gpt-4.1, detector gpt-4o-mini: hint vs plain (conflict)	−3.0 [−6.7, +0.3]	1.000 (0.207)	n.s.	yes
H27	gpt-4.1, detector gpt-6-luna: hint vs plain (conflict)	−3.3 [−7.3, +0.3]	1.000 (0.200)	n.s.	yes
H27	gpt-4.1, detector gpt-6.1-terra: hint vs plain (conflict)	+3.3 [−0.7, +7.0]	1.000 (0.198)	n.s.	yes
H27	gpt-4o-mini, detector claude-haiku-5-5: hint vs plain (conflict)	+6.8 [+2.8, +10.9]	0.027 (0.004)	significant	yes
H27	gpt-4o-mini, detector claude-sonnet-5-5: hint vs plain (conflict)	+10.1 [+6.4, +13.8]	<0.001 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-4.1: hint vs plain (conflict)	+6.8 [+3.7, +10.0]	<0.001 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-4o-mini: hint vs plain (conflict)	+1.3 [−0.4, +3.2]	1.000 (0.294)	n.s.	yes
H27	gpt-4o-mini, detector gpt-6-luna: hint vs plain (conflict)	+5.6 [+2.2, +9.0]	0.036 (0.005)	significant	no
H27	gpt-4o-mini, detector gpt-6.1-terra: hint vs plain (conflict)	+9.4 [+5.9, +13.1]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector claude-haiku-5-5: hint vs plain (conflict)	+9.4 [+5.7, +13.2]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector claude-sonnet-5-5: hint vs plain (conflict)	+14.2 [+10.4, +18.2]	<0.001 (<0.001)	significant	yes

Table 17: Pre-registered tests added in the seventh (§15; H27–H29) (part 2 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	gpt-6-luna, detector gpt-4.1: hint vs plain (conflict)	+7.7 [+4.2, +11.2]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-4o-mini: hint vs plain (conflict)	+2.6 [−1.3, +6.4]	1.000 (0.327)	n.s.	yes
H27	gpt-6-luna, detector gpt-6-luna: hint vs plain (conflict)	+2.9 [+0.2, +5.6]	0.462 (0.077)	n.s.	yes
H27	gpt-6-luna, detector gpt-6.1-terra: hint vs plain (conflict)	+13.1 [+9.4, +16.9]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4.1: hint vs plain (conflict)	−3.3 [−7.7, +1.0]	1.000 (0.242)	n.s.	yes
H27	gpt-6.1-terra, detector gpt-4o-mini: hint vs plain (conflict)	−5.3 [−9.3, −1.3]	0.194 (0.030)	n.s.	yes
H27	gpt-6.1-terra, detector gpt-6-luna: hint vs plain (conflict)	−7.0 [−11.0, −3.3]	0.006 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-6.1-terra: hint vs plain (conflict)	−7.0 [−11.0, −3.0]	0.027 (0.003)	significant	yes
H27	Qwen2.5-7B, detector gpt-4.1: cell vs detector model alone (conflict)	+1.7 [−4.0, +7.3]	1.000 (0.740)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-4o-mini: cell vs detector model alone (conflict)	+6.6 [+0.8, +12.3]	0.312 (0.055)	n.s.	yes
H27	Qwen2.5-7B, detector gpt-6-luna: cell vs detector model alone (conflict)	+8.8 [+3.6, +14.0]	0.021 (0.004)	significant	yes
H27	Qwen2.5-7B, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−14.7 [−20.3, −9.0]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector claude-haiku-5-5: cell vs detector model alone (conflict)	+5.7 [+2.1, +9.3]	0.027 (0.005)	significant	no
H27	claude-haiku-5-5, detector claude-sonnet-5-5: cell vs detector model alone (conflict)	−0.7 [−4.9, +3.6]	1.000 (0.878)	n.s.	yes
H27	claude-haiku-5-5, detector gpt-4.1: cell vs detector model alone (conflict)	+13.1 [+8.1, +18.0]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-4o-mini: cell vs detector model alone (conflict)	+11.6 [+5.4, +17.4]	0.009 (0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6-luna: cell vs detector model alone (conflict)	+18.2 [+13.3, +23.1]	<0.001 (<0.001)	significant	yes
H27	claude-haiku-5-5, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−2.1 [−7.6, +3.3]	1.000 (0.586)	n.s.	yes
H27	gpt-4.1, detector gpt-4.1: cell vs detector model alone (conflict)	+11.7 [+6.7, +16.7]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-4o-mini: cell vs detector model alone (conflict)	+11.2 [+5.2, +17.1]	0.006 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-6-luna: cell vs detector model alone (conflict)	+17.4 [+12.4, +22.3]	<0.001 (<0.001)	significant	yes
H27	gpt-4.1, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−3.3 [−8.7, +2.0]	1.000 (0.379)	n.s.	yes
H27	gpt-4o-mini, detector claude-haiku-5-5: cell vs detector model alone (conflict)	−2.7 [−7.7, +2.3]	1.000 (0.433)	n.s.	yes
H27	gpt-4o-mini, detector claude-sonnet-5-5: cell vs detector model alone (conflict)	−8.7 [−13.7, −3.7]	0.018 (0.003)	significant	no
H27	gpt-4o-mini, detector gpt-4.1: cell vs detector model alone (conflict)	+5.0 [−0.2, +10.1]	0.671 (0.119)	n.s.	yes
H27	gpt-4o-mini, detector gpt-4o-mini: cell vs detector model alone (conflict)	+8.4 [+3.2, +13.8]	0.021 (0.004)	significant	yes

Table 18: Pre-registered tests added in the seventh (§15; H27–H29) (part 3 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H27: the detector model's answer as a hint					
H27	gpt-4o-mini, detector gpt-6-luna: cell vs detector model alone (conflict)	+11.2 [+5.8, +16.7]	0.004 (<0.001)	significant	yes
H27	gpt-4o-mini, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−12.3 [−18.0, −6.7]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector claude-haiku-5-5: cell vs detector model alone (conflict)	+1.1 [−3.8, +5.8]	1.000 (0.785)	n.s.	yes
H27	gpt-6-luna, detector claude-sonnet-5-5: cell vs detector model alone (conflict)	−4.0 [−8.3, +0.3]	0.671 (0.130)	n.s.	yes
H27	gpt-6-luna, detector gpt-4.1: cell vs detector model alone (conflict)	+10.2 [+4.8, +15.6]	0.009 (0.001)	significant	yes
H27	gpt-6-luna, detector gpt-4o-mini: cell vs detector model alone (conflict)	+11.0 [+4.9, +17.0]	0.011 (0.002)	significant	yes
H27	gpt-6-luna, detector gpt-6-luna: cell vs detector model alone (conflict)	+11.1 [+7.3, +15.0]	<0.001 (<0.001)	significant	yes
H27	gpt-6-luna, detector gpt-6.1-terra: cell vs detector model alone (conflict)	−12.1 [−17.2, −6.9]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4.1: cell vs detector model alone (conflict)	+15.7 [+10.7, +20.7]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-4o-mini: cell vs detector model alone (conflict)	+10.2 [+4.2, +16.2]	0.018 (0.003)	significant	yes
H27	gpt-6.1-terra, detector gpt-6-luna: cell vs detector model alone (conflict)	+22.1 [+17.2, +27.0]	<0.001 (<0.001)	significant	yes
H27	gpt-6.1-terra, detector gpt-6.1-terra: cell vs detector model alone (conflict)	+2.7 [−0.3, +5.7]	0.925 (0.200)	n.s.	yes
H28: LLM flags on genuine conflicts					
H28	WikiContradict, gpt-4o-mini: LLM dir. flag vs plain	−6.5 [−10.3, −2.8]	0.001 (0.002)	significant	–
H28	WikiContradict, gpt-4o-mini: LLM removal vs plain	−9.1 [−13.3, −5.1]	<0.001 (<0.001)	significant	–
H28	WikiContradict, gpt-6-luna: LLM dir. flag vs plain	−29.0 [−34.5, −23.6]	<0.001 (<0.001)	significant	–
H28	WikiContradict, gpt-6-luna: LLM removal vs plain	−30.0 [−35.6, −24.5]	<0.001 (<0.001)	significant	–
H28	WikiContradict, claude-haiku-5-5: LLM dir. flag vs plain	−27.7 [−33.2, −22.3]	<0.001 (<0.001)	significant	–
H28	WikiContradict, claude-haiku-5-5: LLM removal vs plain	−28.2 [−33.6, −22.9]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM dir. flag vs plain	−4.0 [−8.4, 0.0]	0.100 (0.157)	n.s.	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM removal vs plain	−21.2 [−29.6, −13.1]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM dir. flag vs plain, gold recall	−3.6 [−6.8, −0.7]	0.024 (0.042)	significant	–
H28	RAMDocs ambiguity, gpt-4o-mini: LLM removal vs plain, gold recall	−17.4 [−22.4, −12.4]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM dir. flag vs plain	−14.8 [−21.5, −8.8]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM removal vs plain	−29.3 [−38.0, −20.9]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM dir. flag vs plain, gold recall	−8.9 [−12.4, −5.7]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6-luna: LLM removal vs plain, gold recall	−20.3 [−25.0, −15.7]	<0.001 (<0.001)	significant	–

Table 19: Pre-registered tests added in the seventh (§15; H27–H29) (part 4 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H28: LLM flags on genuine conflicts					
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM dir. flag vs plain	-5.1 [-10.1, -0.3]	0.068 (0.110)	n.s.	–
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM removal vs plain	-25.6 [-34.0, -17.8]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM dir. flag vs plain, gold recall	-2.7 [-5.8, $+0.2$]	0.089 (0.140)	n.s.	–
H28	RAMDocs ambiguity, claude-haiku-5-5: LLM removal vs plain, gold recall	-16.6 [-21.2, -12.0]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM dir. flag vs plain	-1.0 [-6.1, $+3.0$]	1.000 (1.000)	n.s.	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM removal vs plain	-27.3 [-36.4, -18.2]	<0.001 (<0.001)	significant	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM dir. flag vs plain, gold recall	0.0 [-2.4, $+2.2$]	1.000 (1.000)	n.s.	–
H28	RAMDocs ambiguity, gpt-6.1-terra: LLM removal vs plain, gold recall	-18.5 [-23.6, -13.5]	<0.001 (<0.001)	significant	–
H29: artefact probe of detector direction					
H29	gpt-4o-mini detector, reversed order: direction accuracy vs original	-1.3 [-7.3, $+4.7$]	1.000 (0.810)	n.s.	–
H29	gpt-4o-mini detector, paraphrased: direction accuracy vs original	$+1.0$ [-2.7, $+4.7$]	1.000 (0.805)	n.s.	–
H29	gpt-6-luna detector, reversed order: direction accuracy vs original	-1.7 [-6.0, $+2.7$]	1.000 (0.644)	n.s.	–
H29	gpt-6-luna detector, paraphrased: direction accuracy vs original	-1.3 [-5.7, $+3.0$]	1.000 (0.746)	n.s.	–
H29	gpt-4o-mini detector, reversed order: gpt-4o-mini's gain vs original	-1.9 [-8.4, $+4.7$]	1.000 (0.699)	n.s.	–
H29	gpt-4o-mini detector, paraphrased: gpt-4o-mini's gain vs original	$+1.7$ [-3.1, $+6.3$]	1.000 (0.628)	n.s.	–
H29	gpt-6-luna detector, reversed order: gpt-4o-mini's gain vs original	-4.6 [-8.9, -0.2]	0.196 (0.081)	n.s.	–
H29	gpt-6-luna detector, paraphrased: gpt-4o-mini's gain vs original	-1.2 [-5.1, $+2.7$]	1.000 (0.680)	n.s.	–

Table 20: Pre-registered tests added in the seventh (§15; H27–H29) (part 5 of 5), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H30: the detector's verdict as an answer key					
H30	Qwen2.5-7B, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	+3.0 [−2.0, +8.0]	1.000 (0.398)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+3.0 [−1.7, +7.7]	1.000 (0.355)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	+2.7 [−1.3, +7.0]	1.000 (0.379)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+3.3 [+0.3, +6.3]	1.000 (0.083)	n.s.	yes
H30	claude-haiku-5-5, detector claude-haiku-5-5: LLM dir. flag vs verdict key (conflict)	+0.9 [−1.3, +3.2]	1.000 (0.605)	n.s.	yes
H30	claude-haiku-5-5, detector claude-sonnet-5-5: LLM dir. flag vs verdict key (conflict)	+1.2 [−1.4, +3.9]	1.000 (0.504)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	+1.8 [−1.0, +4.6]	1.000 (0.333)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+1.0 [−1.3, +3.4]	1.000 (0.592)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	+2.1 [−0.4, +4.8]	1.000 (0.209)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+2.7 [+0.6, +4.9]	0.425 (0.035)	n.s.	yes
H30	gpt-4o-mini, detector claude-haiku-5-5: LLM dir. flag vs verdict key (conflict)	+1.4 [−2.0, +4.9]	1.000 (0.549)	n.s.	yes
H30	gpt-4o-mini, detector claude-sonnet-5-5: LLM dir. flag vs verdict key (conflict)	−2.0 [−6.3, +2.2]	1.000 (0.486)	n.s.	yes
H30	gpt-4o-mini, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	+2.1 [−1.7, +5.8]	1.000 (0.398)	n.s.	yes
H30	gpt-4o-mini, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+2.4 [−0.8, +5.8]	1.000 (0.242)	n.s.	yes
H30	gpt-4o-mini, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	+1.4 [−1.9, +4.8]	1.000 (0.541)	n.s.	yes
H30	gpt-4o-mini, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+2.2 [−0.9, +5.3]	1.000 (0.292)	n.s.	yes
H30	gpt-6-luna, detector claude-haiku-5-5: LLM dir. flag vs verdict key (conflict)	−0.8 [−3.0, +1.4]	1.000 (0.655)	n.s.	yes
H30	gpt-6-luna, detector claude-sonnet-5-5: LLM dir. flag vs verdict key (conflict)	−1.2 [−3.1, +0.7]	1.000 (0.354)	n.s.	yes
H30	gpt-6-luna, detector gpt-4.1: LLM dir. flag vs verdict key (conflict)	−1.9 [−3.8, −0.2]	1.000 (0.089)	n.s.	yes
H30	gpt-6-luna, detector gpt-4o-mini: LLM dir. flag vs verdict key (conflict)	+0.2 [−1.0, +1.7]	1.000 (0.937)	n.s.	yes
H30	gpt-6-luna, detector gpt-6-luna: LLM dir. flag vs verdict key (conflict)	−0.7 [−2.7, +1.4]	1.000 (0.699)	n.s.	yes
H30	gpt-6-luna, detector gpt-6.1-terra: LLM dir. flag vs verdict key (conflict)	+0.3 [−2.0, +2.8]	1.000 (0.934)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-4.1: verdict key vs hint (conflict)	+5.7 [+0.3, +10.7]	0.393 (0.073)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-4o-mini: verdict key vs hint (conflict)	+1.3 [−4.3, +6.7]	1.000 (0.810)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6-luna: verdict key vs hint (conflict)	+1.7 [−3.7, +6.7]	1.000 (0.709)	n.s.	yes
H30	Qwen2.5-7B, detector gpt-6.1-terra: verdict key vs hint (conflict)	−7.3 [−12.7, −2.3]	0.095 (0.015)	n.s.	yes

Table 21: Pre-registered tests added in the eighth (§16; H30–H31) (part 1 of 3), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H30: the detector's verdict as an answer key					
H30	claude-haiku-5-5, detector claude-haiku-5-5: verdict key vs hint (conflict)	+10.2 [+6.3, +14.2]	<0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector claude-sonnet-5-5: verdict key vs hint (conflict)	+13.3 [+8.8, +17.9]	<0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector gpt-4.1: verdict key vs hint (conflict)	+12.3 [+8.6, +16.2]	<0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector gpt-4o-mini: verdict key vs hint (conflict)	−2.4 [−6.8, +1.9]	1.000 (0.402)	n.s.	yes
H30	claude-haiku-5-5, detector gpt-6-luna: verdict key vs hint (conflict)	+8.2 [+4.3, +12.1]	0.001 (<0.001)	significant	yes
H30	claude-haiku-5-5, detector gpt-6.1-terra: verdict key vs hint (conflict)	+8.3 [+4.6, +12.2]	<0.001 (<0.001)	significant	yes
H30	gpt-4o-mini, detector claude-haiku-5-5: verdict key vs hint (conflict)	+4.8 [+0.1, +9.3]	0.393 (0.083)	n.s.	yes
H30	gpt-4o-mini, detector claude-sonnet-5-5: verdict key vs hint (conflict)	+7.8 [+3.2, +12.3]	0.025 (0.004)	significant	yes
H30	gpt-4o-mini, detector gpt-4.1: verdict key vs hint (conflict)	+9.7 [+5.1, +14.2]	0.005 (<0.001)	significant	yes
H30	gpt-4o-mini, detector gpt-4o-mini: verdict key vs hint (conflict)	+4.7 [−0.7, +9.9]	0.440 (0.139)	n.s.	yes
H30	gpt-4o-mini, detector gpt-6-luna: verdict key vs hint (conflict)	+6.7 [+2.0, +11.3]	0.075 (0.011)	n.s.	no
H30	gpt-4o-mini, detector gpt-6.1-terra: verdict key vs hint (conflict)	−2.1 [−6.7, +2.4]	1.000 (0.502)	n.s.	yes
H30	gpt-6-luna, detector claude-haiku-5-5: verdict key vs hint (conflict)	+5.7 [+0.4, +10.9]	0.388 (0.067)	n.s.	yes
H30	gpt-6-luna, detector claude-sonnet-5-5: verdict key vs hint (conflict)	+5.1 [+0.2, +10.0]	0.393 (0.080)	n.s.	yes
H30	gpt-6-luna, detector gpt-4.1: verdict key vs hint (conflict)	+15.6 [+10.8, +20.3]	<0.001 (<0.001)	significant	yes
H30	gpt-6-luna, detector gpt-4o-mini: verdict key vs hint (conflict)	+5.8 [+0.2, +11.4]	0.393 (0.083)	n.s.	yes
H30	gpt-6-luna, detector gpt-6-luna: verdict key vs hint (conflict)	+8.9 [+4.1, +13.7]	0.006 (<0.001)	significant	yes
H30	gpt-6-luna, detector gpt-6.1-terra: verdict key vs hint (conflict)	−6.1 [−11.0, −1.2]	0.209 (0.035)	n.s.	yes
H31: randomised direction					
H31	gpt-4o-mini: right-flag effect (conflict)	+29.4 [+22.6, +36.3]	<0.001 (<0.001)	supported	–
H31	gpt-6-luna: right-flag effect (conflict)	+33.5 [+26.8, +40.3]	<0.001 (<0.001)	supported	–
H31	claude-haiku-5-5: right-flag effect (conflict)	+24.1 [+18.0, +30.2]	<0.001 (<0.001)	supported	–
H31	gpt-6.1-terra: right-flag effect (conflict)	+17.6 [+11.3, +23.9]	<0.001 (<0.001)	supported	–
H31	gpt-4o-mini: wrong-flag effect (conflict)	−31.2 [−39.0, −23.4]	<0.001 (<0.001)	supported	–
H31	gpt-6-luna: wrong-flag effect (conflict)	−44.9 [−52.0, −37.8]	<0.001 (<0.001)	supported	–
H31	claude-haiku-5-5: wrong-flag effect (conflict)	−49.2 [−56.5, −42.1]	<0.001 (<0.001)	supported	–
H31	gpt-6.1-terra: wrong-flag effect (conflict)	−64.5 [−72.3, −56.0]	<0.001 (<0.001)	supported	–

Table 22: Pre-registered tests added in the eighth (§16; H30–H31) (part 2 of 3), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

from a different model family, validated on about 100 human labels. Neither the judge nor the labels were available, so CONFLICTS is scored only on items with a reference answer, with one seed, and reported as exploratory.

4. *H4.* The pre-registration counted 54 false-flag cells. On clean questions, cells that differ only in direction accuracy render identical prompts (their scores differ by exactly 0), so they are one comparison: 18 in all, 9 per format. We report H4a per format. The cell-permutation p-value for the dose-response correlation assumed exchangeable cells; the cells are nested, so we replace it with a question-level bootstrap CI. No threshold was set for H4c, so we report a count.

5. *H5.* A non-significant difference is not equivalence. We add a TOST with a ± 3-point margin, chosen in the addendum after the confirmatory run, so it is post hoc.

6. *H6.* The first draft tested whether the grid prediction lies inside the actual effect's CI, which is lenient. We now report the pre-registered test (actual effect inside a bootstrap prediction interval) and a paired test of the two spaces' errors.

7. *Placement modes.* For H6, the detectors without direction were run under the label `remove_suspect` as well as `remove`. Both render identical prompts, so these were served entirely from the response cache.

8. *Significance.* Earlier drafts used different corrections in different tables. We now apply one scheme throughout (Section 4), which makes some exploratory differences that were bold in earlier drafts non-significant.

9. *Single- and multi-answer split.* The split of conflict questions by number of gold answers (Appendix C.1) was chosen after seeing the minimal-skeleton run and is exploratory.

10. *Second registration.* We later registered H12–H14 (ConflictBank, the neutral flag with a resolution instruction, and the published baselines), the Qwen2.5-7B detector and the ConflictBank sampling rule. The gpt-4.1 runs in it use seed 0 only, to stay within budget. One ConflictBank gpt-4o-mini run stopped on a transient API error and was resumed from the response cache; no row was re-sampled.

11. *Third and fourth registrations.* H15–H18 (§11) and H19 (§12) were committed before their runs. To stay within budget, §12 dropped two §11 runs before they started: gpt-4.1 under majority misinformation (replaced by gpt-6.1-terra in H17) and gpt-4.1 seeds 1–2, so H14 with gpt-4.1 stays on seed 0 as first registered and every gpt-4.1 baseline comparison is seed-0 matched.

12. *Reasoning models.* gpt-6-luna and gpt-6.1-terra accept only their default temperature and require `max_completion_tokens`; we used their defaults with a 4,096-token budget. 25 responses from the two models (almost all closed-book) were empty because reasoning used the whole budget; they are scored as wrong.

13. *Fifth and sixth registrations.* H20–H24 (§13) and H25–H26 with Claude and RAGuard (§14) were committed before their runs. The planned `cross_mj` runs completed within budget. Gemini was planned but not run, because no API key was available. Claude rejects a temperature and JSON mode on the endpoint we used; it ran at its default temperature, and as a detector its JSON is parsed by the detector's regex fallback. The answer parser unwraps paired tags (`<answer>X</answer>`), which no earlier result contained.

14. *Permutation resolution.* With 10,000 permutations the smallest p-value is 10^{-4}, which cannot survive Holm over 636 exploratory comparisons. After the §14 runs, and before writing them up, we extended the permutations to 200,000 whenever none of the first 10,000 reached the observed statistic. This changes only p-values at their floor, and only lowers them.

15. *H24 additions.* The cross-cell break-even point, the count of positive cells and the mechanism score (credit for naming the false side where the generator errs, charge for marking a correct passage where it is right) were defined after seeing the H24 cells and are exploratory.

16. *Seventh registration.* H27–H29 and the reanalyses of §15 were committed before their runs. To finish within the session, the §15 runs were started in parallel tracks rather than strictly in priority order; spend never approached the $195 limit, so no run was dropped. The question-level model adds one dependency (statsmodels 0.14.5). H21(c) is now the ratio of correct to answered answers; the registered

per-question mean averaged the two conditions over different questions (those not abstained in every seed), which is why an earlier draft's selective accuracies did not match Table 1. Both are in the released statistics.

17. *Eighth registration.* H30–H31 and the re-analyses of §16 were committed before their runs, which ran in parallel tracks within the budget. The pooled estimate now uses a multilevel model in place of the DerSimonian–Laird bootstrap, the moderators were recoded (answer policy; natural misinformation is RAGuard only), and the utility analysis uses the same settings as the pooled estimate. "Following" a flag now has one definition for right and wrong flags (§16.3); an earlier draft compared correct answers with any wrong answer.

18. *Run interruptions.* Several runs were stopped and resumed (a transient API error; reordering the run queue). Resumption keeps completed rows and re-requests only missing ones; no row was re-sampled.

The addenda (H10–H31 and the re-analyses above) and the Qwen seed change were each committed to the repository before the runs they cover. The repository's commit history is the only timestamp; we did not use a third-party registry (van Miltenburg et al., 2021).

RAMDocs labels. The released document types are imperfect: 163 documents typed *correct* contain a wrong answer, some as injected misinformation, others incidentally. We leave pairs that touch them out of pair-level metrics, and questions whose only sign of conflict is such a document (55 of 500; 35 of the confirmatory split) count only in "All". Treating them as misinformation is a robustness check.

Power. With 139 RAMDocs conflict questions, the minimum detectable effect at 80% power is 4–12 points (median 5.8; Card et al., 2020).

Metric choice. With the one-way metric, our port of MADAM-RAG scores 28.5% on the pilot split and 32.8% on all 500 questions (published: 28.00, on an undisclosed subset); with the bidirectional metric it scores 38.4% on all 500. Because the published subset and scorer are unknown, we do not claim to reproduce the published number. Normalisation lower-cases and removes punctuation and articles.

WikiContradict and CONFLICTS. A WikiContradict answer is correct if both of the dataset's answer strings (one per passage) match a final answer under the bidirectional rule. From CONFLICTS we use the 237 questions that have a reference answer (outdated-information, misinformation and no-conflict items); an answer is correct if it contains the reference answer.

Baselines. MADAM-RAG is ported unchanged to our client: three debate rounds, temperature 0, 128 tokens per call. Astute RAG has no official code. We use the prompts in its appendix verbatim with its default settings (t=1 consolidation round, m=1 generated passage, temperature 0, 1,024 output tokens). A generated passage that says "I don't know" is dropped, and the answer is the text in its `<ANSWER>` tag. TrustRAG is ported from its repository (commit `11dcea0`, MIT licence): SimCSE CLS embeddings (`sup-simcse-bert-base-uncased`), a ROUGE-L gate of 0.25, 2-means clustering and a cosine threshold of 0.88, followed by its three prompts verbatim at temperature 0.1. Its system message is not sent, because our client sends user messages only. The *multi* variants change one sentence of the final step to ask for every valid answer (Astute RAG) or append our final-answer line (TrustRAG); the unmodified variants are also reported.

ConflictBank. We use the QA file of the dataset's Hugging Face release (revision `92866a0`). Questions are sampled round-robin over relations with seed 0; distractors are other subjects' correct passages. The sampling script, the item ids and the seed are released; the data are not redistributed, because the release states a licence in its README but ships no licence file.

RAGuard. We use the Hugging Face release `UCSC-IRKM/RAGuard` (revision `dae344a2`, MIT licence on the card). A post is usable if it has at least 15 words after removing the "[Link Post]" placeholder and is not a duplicate within its claim; posts are cut to 300 words and shuffled by claim id. Supporting × misleading pairs are the contradictions and misleading posts the misinformation side. The build script and item ids are released.

Cost and compute. All API calls in this paper, pilots included, cost $187.38. Prices were checked on 2026-10-27 (per million input, cached-input and output tokens: gpt-4o-mini $0.15, $0.075

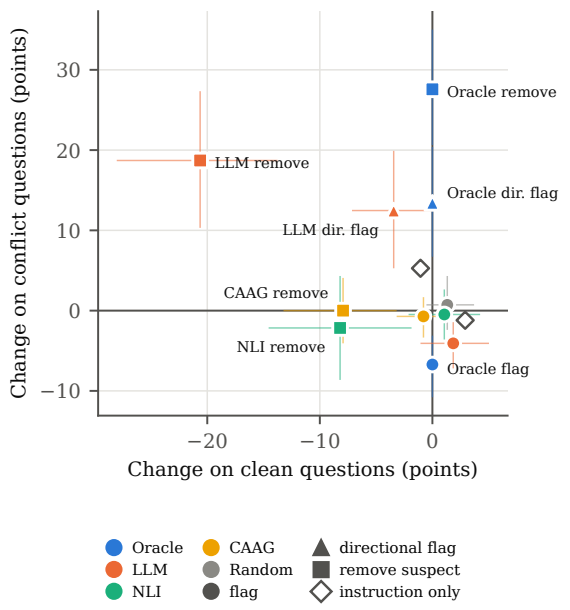


Figure 4: Change in one-way strict accuracy on RAM-Docs conflict questions (y) against clean questions (x), relative to no flags (gpt-4o-mini, default skeleton, confirmatory split, 95% CIs). Colour: detector; shape: format. The oracle sits on the y-axis because it never flags a clean question. Without direction, removal deletes the second passage of each pair.

and $0.60; gpt-4.1 $2.00, $0.50 and $8.00; gpt-6-luna $0.10, $0.01 and $0.50; gpt-6.1-terra $1.00, $0.05 and $5.00; Claude Haiku 5.5 $0.10, $0.01 and $0.50; Claude Sonnet 5.5 $2.00, $0.10 and $10.00). CAAG and NLI ran on a laptop CPU (0.2 s and 1.2 s per question).

E Synthetic Detectors and the Grid

Construction. Each synthetic detector (Section 3.3) is built from the gold labels. Each gold contradiction pair is flagged with probability r. False flags are drawn from non-contradicting pairs at a single per-pair rate, set over the whole split so that expected precision is p. That rate is dataset-wide, so conflict-free questions receive false flags too. A true flag names the gold misinformation side with probability d, and otherwise the other side. A false flag has no misinformation side, so its suspect is picked uniformly. Each pair gets one uniform draw for flagging and one for direction, and every cell reuses them. Cells are therefore nested: raising r or d, or lowering p, only adds flags or corrections. This keeps comparisons across cells paired.

Results. Figure 5 replaces the oracle with synthetic detectors of known quality. Gains rise with direction accuracy in both formats. Only the cells with perfect direction are significant on all questions (directional flag +6, removal +13 points at perfect precision), but neighbouring cells at direction accuracy 0.75 have similar point estimates on conflict questions, so this is a dose-response, not a threshold: with 36 cells under Holm, the grid has little power to separate adjacent levels. Precision matters mainly for removal. On clean questions a cell's prompt does not depend on direction accuracy, so the false-flag cells reduce to 9 distinct comparisons per format. Removal is significantly harmful in 6 of them, the directional flag in 0 (H4a): false flags make the model delete correct passages, which costs clean questions 15 points at precision 0.5 and 26 at 0.25, while synthetic false directional flags cost at most two points. The harm tracks the share of clean questions that lose a correct document (Spearman ρ =-0.39 over 18 distinct cells, 95% CI [−0.74, −0.12] from a question-level bootstrap; H4b). Removal hurt clean questions significantly more than the directional flag in 5 of 9 matched cells (H4c; we set no threshold, so we report the count). Removal is the more powerful and the more fragile interface: its best cell gains twice the directional flag's on all questions, but below precision 0.75 it is the only format that does net harm.

Placing real detectors on the grid. Can the grid predict a real detector's effect from a few summary numbers? We compare two ways of summarising a detector. *Pair-level*: its pair precision, recall and direction accuracy. *Document-level*: what it does to each subset's documents in a given format. For conflict questions, that is the share whose misinformation documents are all marked and the share that have a correct document marked; for clean questions, the share that have a correct document marked. Here "marked" means removed (Remove) or headed as misinformation (Directional flag). The prediction for a subset is the inverse-distance-weighted mean effect of the four nearest grid cells in that space.

Does the grid predict real detectors? If the grid only described synthetic detectors, it would be of limited use. Figure 2 (Appendix B) tests whether it predicts real ones (H6). As preregistered, we bootstrap the whole prediction (cell effects, every detector's location and the interpo-

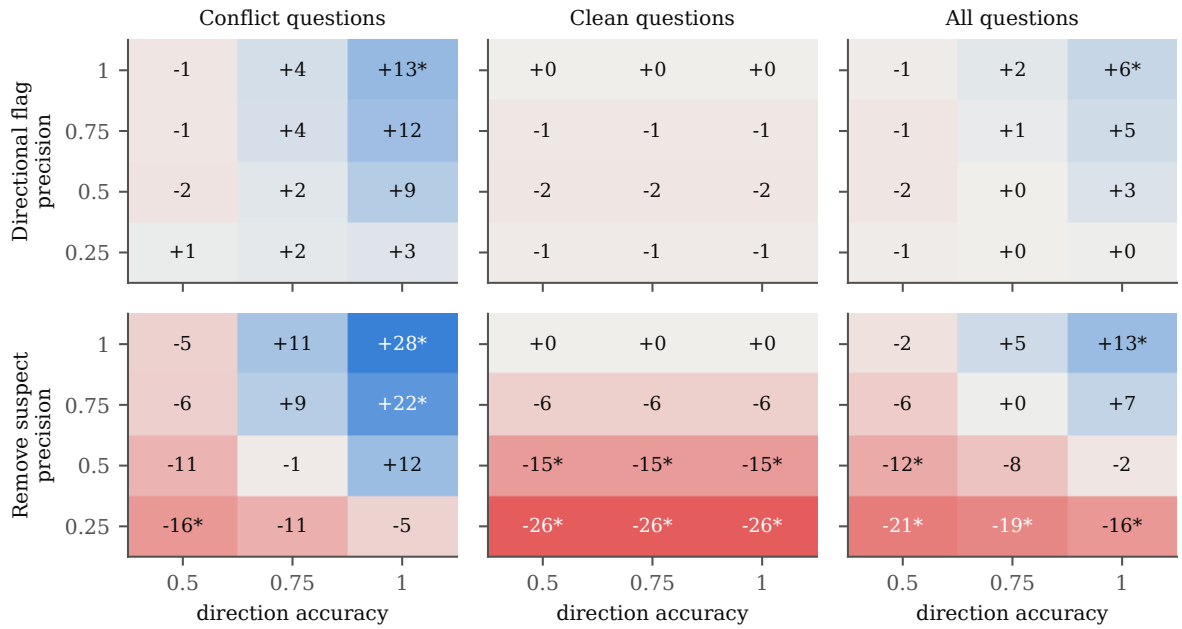


Figure 5: Synthetic detectors at recall 1: change in one-way strict accuracy (points) against no flags, by pair precision (rows) and direction accuracy (columns), for the directional flag (top) and removal (bottom). *: significant after Holm correction over the 36 cells of that mode and subset. Clean questions only see false flags, so their rows do not vary with direction accuracy.

lation) over questions, and ask whether the actual effect falls inside the 95% interval. Locating detectors by what they do to documents (the share of conflict questions whose misinformation is marked, and the share of questions that have a correct document marked) puts 7 of 8 effects inside; locating them by pair precision, recall and direction accuracy puts 6 inside. Document-level errors are smaller on average (4.1 vs. 5.3 points), but the paired difference, -1.2 [-3.6, $+0.4$] (exact sign-flip $p = 0.41$), is not significant over 8 cases. Both spaces miss the same case, the LLM detector's removal gain on conflict questions: one correctly directed pair is enough to remove a misinformation document, so the detector removes the misinformation on 61.2% of conflict questions despite its low pair recall. We chose the document-level axes after seeing the LLM detector on the pilot split. The grid is a useful first estimate of a detector's effect, but with three detectors we cannot claim that one summary predicts better than the other.

	Pre-registered test	Estimate [95% CI]	p_{Holm} (q_{BH})	Outcome	Robust
H31: randomised direction					
H31	gpt-4o-mini: right-flag effect + wrong-flag effect (conflict)	-1.9 [-12.2, $+8.6$]	0.733 (0.814)	n.s.	–
H31	gpt-6-luna: right-flag effect + wrong-flag effect (conflict)	-11.4 [-21.0, -1.8]	0.049 (0.043)	significant	–
H31	claude-haiku-5-5: right-flag effect + wrong-flag effect (conflict)	-25.1 [-34.3, -15.8]	$<$0.001 ($<$0.001)	significant	–
H31	gpt-6.1-terra: right-flag effect + wrong-flag effect (conflict)	-46.9 [-57.1, -36.4]	$<$0.001 ($<$0.001)	significant	–

Table 23: Pre-registered tests added in the eighth (§16; H30–H31) (part 3 of 3), each committed before the runs it covers. Same conventions as Table 6. Settings name the benchmark and generator. ConflictBank rows from H15 on use the single-answer prompt unless marked *forced*; *CB majority* has three wrong-side passages per conflict question; RAGuard's misleading posts are real Reddit text.

	Effect (points)			Follow rate (%)		Share of headroom (%)		Net at direction precision		
Generator	Right	Wrong	Break-even [95% CI]	Right	Wrong	Right	Wrong	0.50	68.8%	82.6%
gpt-4o-mini	$+29.4$	-31.2	0.52 [0.43, 0.60]	57 [46, 67]	51 [40, 62]	59	59	-0.9	$+10.5$	$+18.8$
gpt-6-luna	$+33.5$	-44.9	0.57 [0.51, 0.64]	72 [60, 83]	77 [67, 86]	76	87	-5.7	$+9.1$	$+19.9$
Claude Haiku	$+24.1$	-49.2	0.67 [0.61, 0.74]	63 [49, 76]	64 [55, 73]	70	72	-12.5	$+1.3$	$+11.4$
gpt-6.1-terra	$+17.6$	-64.5	0.79 [0.72, 0.85]	67 [53, 80]	95 [89, 99]	62	96	-23.5	-8.0	$+3.4$

Table 24: Randomised direction (H31; ConflictBank, one-answer prompt, conflict questions), re-analysed (exploratory). *Break-even*: the direction precision at which a flag on the true pair neither helps nor hurts, $-W/(R-W)$ for right-flag effect R and wrong-flag effect W; bootstrap CI within arms. *Follow rate*: a right flag is followed when an answer that was wrong without the flag becomes correct, a wrong flag when a correct answer comes to contain the misinformation; question-bootstrap CIs. *Share of headroom*: R over the no-flag error rate of the right arm, and $-W$ over the no-flag accuracy of the wrong arm. *Net*: $pR + (1-p)W$ at direction precision p: 0.50, gpt-4o-mini's on ConflictBank, and the most precise detector's (gpt-6.1-terra). These nets assume the detector flags the true pair; real detectors also flag other pairs and clean questions. Figure 1 (right) uses the follow rates to predict the 30 crossed cells.

Setting	Conflict	Clean	5%	20%	Break-even
CB multi-answer, Qwen2.5-7B	+6.7	−1.0	−0.6	+0.5	13
CB multi-answer, gpt-4.1	+17.0	0.0	+0.9	+3.4	–
CB multi-answer, gpt-4o-mini	+20.0	−1.0	+0.1	+3.2	5
CB forced, Claude Haiku	−4.7	−1.3	−1.5	−2.0	–
CB forced, gpt-4o-mini	+4.0	−2.7	−2.3	−1.3	40
CB forced, gpt-6-luna	−2.4	−1.3	−1.4	−1.6	–
CB forced, gpt-6.1-terra	−14.3	0.0	−0.7	−2.9	–
CB majority, Claude Haiku	+6.8	−5.0	−4.4	−2.6	42
CB majority, gpt-4o-mini	+5.7	−4.7	−4.2	−2.6	45
CB majority, gpt-6-luna	+7.1	−1.3	−0.9	+0.4	16
CB majority, gpt-6.1-terra	+9.0	0.0	+0.4	+1.8	–
CB one-answer, Qwen2.5-7B	+7.3	−1.0	−0.6	+0.7	12
CB one-answer, Claude Haiku	−4.1	−5.0	−5.0	−4.8	–
CB one-answer, Claude Sonnet	−2.7	−1.0	−1.1	−1.3	–
CB one-answer, gpt-4.1	−2.3	−3.0	−3.0	−2.9	–
CB one-answer, gpt-4o-mini	+8.4	−4.7	−4.0	−2.0	36
CB one-answer, gpt-6-luna	+8.6	−1.3	−0.8	+0.6	13
CB one-answer, gpt-6.1-terra	−11.7	0.0	−0.6	−2.3	–
RAMDocs, Qwen2.5-7B (multi-answer)	+2.9	−7.1	−6.6	−5.1	71
RAMDocs, Claude Haiku	+22.5	−4.5	−3.1	+0.9	17
RAMDocs, Claude Sonnet	+12.2	−3.2	−2.4	−0.1	21
RAMDocs, gpt-4.1 (multi-answer)	+8.4	0.0	+0.4	+1.7	–
RAMDocs, gpt-4o-mini (multi-answer)	+12.5	−3.4	−2.6	−0.3	22
RAMDocs, gpt-4o-mini (minimal)	+3.6	−10.3	−9.6	−7.5	74
RAMDocs, gpt-6-luna	+29.7	−11.6	−9.6	−3.4	28
RAMDocs, gpt-6.1-terra	+38.8	−0.8	+1.2	+7.1	2
RAGuard, Claude Haiku	+5.0	+1.3	+1.5	+2.1	–
RAGuard, Claude Sonnet	+7.1	+10.0	+9.9	+9.4	–
RAGuard, gpt-4o-mini	+0.6	+1.7	+1.6	+1.4	–
RAGuard, gpt-6-luna	+3.1	−1.3	−1.1	−0.5	30
RAGuard, gpt-6.1-terra	+2.0	+1.0	+1.1	+1.2	–

Table 25: The gpt-4o-mini LLM detector's directional flag: effect (points, against no flags) on conflict and clean questions, and reweighted to a 5% and a 20% share of conflict questions (exploratory). *Break-even*: the share of conflict questions (%) above which the flag helps, where it hurts on clean questions. Oracle flags never fire on clean questions, so their reweighted effect is their conflict effect times the share.

		Plain	Oracle, conflict q.			LLM, directional flag		LLM, remove suspect	
Generator	Skeleton	conflict	flag	dir. flag	remove	conflict	clean	conflict	clean
gpt-4o-mini	default	22.1	**−6.7**	**+13.4**	**+27.6**	+12.5	−3.4	+18.7	**−20.6**
gpt-4o-mini	minimal	20.4	**−5.3**	+5.0	**+18.9**	+3.6	−10.3	+11.0	**−19.6**
gpt-4.1	default	21.8	**−8.6**	**+18.2**	**+30.2**	+8.4	0.0	**+20.6**	**−24.3**
gpt-4.1	minimal	33.6	**−11.0**	**+12.7**	**+14.9**	–	–	–	–
Qwen2.5-7B	default	19.4	−5.8	+3.6	**+18.0**	+2.9	−7.1	+7.9	**−18.3**
Qwen2.5-7B	minimal	17.3	−5.8	+2.9	**+10.8**	–	–	–	–
gpt-6-luna‡	default	14.4	**−3.1**	**+39.1**	**+38.8**	**+29.7**	**−11.6**	–	–
gpt-6.1-terra‡	default	10.8	−0.7	**+39.6**	**+39.6**	**+38.8**	−0.8	–	–

Table 26: The main effects across generators and prompt skeletons (RAMDocs hold-out, 3 seeds, one-way strict accuracy). *Plain*: accuracy (%) on conflict questions with no flags. Other columns: paired difference from plain in points. *Default* skeleton tells the model to give every valid answer; *minimal* drops that policy (Appendix A). **Bold**: significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). The LLM detector is gpt-4o-mini for every generator. Qwen2.5-7B-Instruct runs locally (4-bit), seed 0. ‡: current reasoning models at their default temperature (gpt-6-luna 3 seeds, gpt-6.1-terra seed 0).

	Pair level			Clean-q.	Direc-	Removal, per question		Latency	Cost
Detector	P	R	F1	flagged	tion	misinfo gone	correct lost	(s)	($/1k q.)
CAAG (heuristic)	21.7	7.9	11.6	34.1	–	9.4	31.7	0.23	0.000
NLI cross-encoder	20.1	29.3	23.8	61.9	–	44.6	52.4	1.18	0.000
LLM (gpt-4o-mini)	37.2	20.5	26.4	76.2	88.3	61.2	65.9	0.80	0.181

Table 27: Detectors on the RAMDocs hold-out split (300 questions; percentages). *Clean-q. flagged*: share of conflict-free questions with at least one flag. *Direction*: share of correctly flagged contradictions whose suspect side is the misinformation. *Misinfo gone*: conflict questions whose misinformation documents are all removed when removing the suspect (or, without direction, the second passage). *Correct lost*: clean questions that lose at least one correct document. Latency is per question; CAAG and NLI run locally.

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	35.2	37.2	25.9	49.7	911
Prompt only					
Instruction (v1)	+1.3	+0.5	−1.0	+2.1	948
Instruction (v2)	+2.2	+1.4	+3.6	−1.1	951
Oracle detector					
Oracle, flag	−3.6	−4.0	**−7.7**	0.0	980
Oracle, directional flag	+7.0	**+7.9**	**+15.1**	0.0	998
Oracle, remove suspect	**+15.4**	**+17.5**	**+33.3**	0.0	825
Real detectors (exploratory)					
Random, flag	+0.4	+0.4	−0.2	+1.1	949
CAAG, flag	−0.9	−1.4	−1.9	−0.8	942
CAAG, remove	−4.0	−4.5	−1.0	−8.5	809
NLI, flag	+0.2	0.0	−1.4	+1.6	1021
NLI, remove	−8.0	−7.9	−3.1	−13.2	592
LLM, flag	−3.4	−3.4	−7.2	+0.8	970
LLM, directional flag	+4.7	+4.5	+14.1	−6.1	983
LLM, remove suspect	−2.1	−2.6	+18.9	**−26.5**	718
Reference					
MADAM-RAG	+5.7	+5.2	+3.6	+6.9	8196

Table 28: RAMDocs hold-out, gpt-4o-mini, 3 seeds, bidirectional strict accuracy (robustness). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant under the paper's single scheme (Section 4: a registered test's Holm-adjusted p within its family, otherwise Holm over all exploratory comparisons). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

	Test (labels: *contains*)	One-way [95% CI]	p_{Holm}	Bidir.	p_{Holm}
H2	oracle+flag < plain (conflict)	−4.6 [−8.1, −1.5]	0.006	−5.2	0.003
H3	oracle+flag_directional > plain (conflict)	+11.6 [+5.6, +17.3]	<0.001	+13.3	<0.001
H3	oracle+flag_directional > oracle+flag (conflict)	+16.2 [+10.8, +21.8]	<0.001	+18.5	<0.001
H3	oracle+remove_suspect > plain (conflict)	+25.8 [+19.5, +32.2]	<0.001	+30.3	<0.001
H4	harmful cells, clean q. (dir. flag; removal); ρ	0/9; 8/9; $\rho = -0.08$ [-0.56, -0.02]			

Table 29: Label robustness: pre-registered tests with every suspect document treated as misinformation (173 conflict and 127 clean hold-out questions; this rule assigns the 35 questions the main analysis excludes). The bootstrap distribution of the H4 correlation is skewed, so its point estimate lies near the end of its percentile interval.

CAAG setting	θ_s	θ_c	P	R	F1	Clean FPR
Default	0.65	0.35	21.7	7.9	11.6	34.1
Best pair F1	0.40	0.10	20.7	49.6	29.2	87.3
Best question F1	0.55	0.15	22.2	28.8	25.1	60.3

Signals (defaults)	Pair F1	Flag rate	Clean FPR
all signals	11.6	6.2	34.1
negation only	0.6	0.4	4.0
entity only	8.6	4.3	23.0
temporal only	0.0	0.0	0.0

Table 30: CAAG on the RAMDocs hold-out (%). Top: thresholds from a sweep over 121 settings ($\theta_s \in [0.40, 0.90]$, $\theta_c \in [0.10, 0.60]$). Bottom: each signal alone.

Condition	All	Lab.	Conflict	Clean
No flags (plain)	32.7	34.5	21.8	48.4
Instruction (v2)	+5.8	+5.2	+12.2	−2.6
Oracle, flag	−4.0	−4.5	**−8.6**	0.0
Oracle, directional flag	**+8.4**	**+9.6**	**+18.2**	0.0
Oracle, remove suspect	**+14.0**	**+15.8**	**+30.2**	0.0
LLM, directional flag	+5.4	+4.4	+8.4	0.0
LLM, remove suspect	+1.0	−0.8	**+20.6**	**−24.3**
NLI, remove	−7.0	−7.3	+0.2	**−15.6**

Table 32: Replication with gpt-4.1 as generator (RAMDocs hold-out, 3 seeds, one-way strict accuracy; detector fixed to gpt-4o-mini). Same conventions as Table 5.

Condition	All	Conflict	No conflict
No flags (plain)	65.8	46.9	73.6
Instruction (v2)	+0.6	+3.9	−0.8
CAAG, flag	+1.1	+2.4	+0.6
NLI, flag	−0.3	−0.5	−0.2
LLM, flag	+2.7	+5.3	+1.6
LLM, directional flag	+0.3	+1.4	−0.2

Table 31: CONFLICTS, 237 automatically scorable questions (69 conflict), gpt-4o-mini, 3 seeds; exploratory. Contains-match accuracy (%) and differences in points; bold would mark Holm-adjusted $p < 0.05$ within this table, and no difference reaches it. CAAG: flags 94.2% of conflict and 92.9% of clean questions; NLI: flags 52.2% of conflict and 43.5% of clean questions; LLM: flags 98.6% of conflict and 83.9% of clean questions.

Condition	gpt-4o-mini	gpt-4.1
No flags (plain)	14.0	29.0
Instruction (v1)	**+8.7**	–
Instruction (v2)	**−6.7**	–
Oracle, flag	**+6.5**	**+19.0**
CAAG, flag	+0.7	–
NLI, flag	+3.2	–
LLM, flag	**+6.2**	–

Table 33: WikiContradict (253 genuine conflicts between two Wikipedia passages; the reference answer is both sides), 3 seeds (gpt-4.1: the oracle flag only). First row: share of answers giving both sides (%); other rows: difference in points, **bold** = significant under the paper's single scheme (H8 is a registered family; the gpt-4.1 column is exploratory).

	RAMDocs, gpt-4o-mini			RAMDocs, gpt-4.1		ConflictBank		CB majority		
Method	Lab.	Conflict	Clean	Lab.	Conflict	Conflict	Clean	Conflict	Clean	Calls
No flags (plain), accuracy	31.3	22.1	41.5	32.8†	19.4†	51.1	89.0	21.6	89.0	1
Instruction (v2) only	+2.3	+5.3	−1.1	–	+12.2	–	–	–	–	1
Oracle, flag + instr.	−1.8	−2.4	−1.1	+1.5†	+5.8†	–	–	–	–	1
Oracle, dir. flag + instr.	+6.4	+13.2	−1.1	**+14.7**†	**+30.9**†	–	–	–	–	1
MADAM-RAG	+5.7	+3.6	+7.9	–	–	–	–	–	–	9×tok.
Astute RAG	**−15.6**	−8.4	**−23.5**	–	–	−2.1	−9.7	+1.4	−9.7	2
Astute RAG (multi)	−5.9	−1.2	−11.1	+2.3†	+13.7†	–	–	–	–	2
TrustRAG	**−17.5**	−11.3	**−24.3**	–	–	−9.7	−4.0	+6.8	−4.0	3
TrustRAG (multi)	**−18.1**	−12.2	**−24.6**	−10.9†	0.0†	–	–	–	–	3
Single call (name suspects, then answer)	+6.2	+12.0	−0.3	–	–	+7.6	−1.0	+6.8	−1.0	1
LLM detector, flag + instr.	−2.0	−2.2	−1.9	–	–	–	–	–	–	1+1
LLM detector, dir. flag + instr.	−3.5	+8.4	**−16.7**	–	–	–	–	–	–	1+1
LLM detector, dir. flag	+4.9	+12.5	−3.4	+8.7†	+14.4†	+8.4	−4.7	**+5.7**	−4.7	1+1
LLM detector, remove suspect	0.0	+18.7	**−20.6**	–	**+20.6**	+7.9	−7.0	+7.0	−7.0	1+1

Table 34: Resolution instructions, published baselines and a single-call baseline. First row: accuracy (%, one-way strict) with no flags; other rows: paired difference from it in points. RAMDocs uses the multi-answer prompt (it has same-name questions with several answers); ConflictBank and its majority variant (three wrong-side passages, one correct) use the single-answer prompt and gpt-4o-mini. *Lab.*: labelled RAMDocs questions. 3 seeds, except †: gpt-4.1 seed 0 only, relative to the seed-0 no-flag accuracy in the first row (other gpt-4.1 cells are relative to its 3-seed no-flag accuracy, Table 32). *instr.*: the v2 reliability instruction. *(multi)*: the baseline's final step asks for every valid answer. *Single call*: one call that names suspect passages, then answers. Calls per question; *1+1*: one detector call and one generator call. **Bold**: significant under the paper's single scheme.

		Detector					Accuracy with				Flag minus	
Detector	Generator	Right	Wrong	Gen. acc.	Score	Gain	flag	verdict	hint	det. alone	hint	det. alone
gpt-4o-mini	gpt-6.1-terra	67.0	30.3	73.0	−4.7	**−11.7**	61.3	–	67.7	51.1	−6.3	**+10.2**
gpt-4o-mini	Claude Haiku	67.0	30.3	66.8	+1.9	−4.1	62.7	61.7	64.1	51.1	−1.4	**+11.6**
gpt-6.1-terra	gpt-6.1-terra	29.7	4.3	73.0	+2.7	+2.7	75.7	–	66.0	73.0	**+9.7**	+2.7
gpt-4o-mini	gpt-4.1	67.0	30.3	64.7	+4.0	−2.3	62.3	–	61.7	51.1	+0.7	**+11.2**
gpt-6.1-terra	Claude Haiku	29.7	4.3	66.8	+4.3	**+4.1**	70.9	68.2	59.9	73.0	**+11.0**	−2.1
gpt-6.1-terra	gpt-4.1	29.7	4.3	64.7	+4.7	**+5.0**	69.7	–	68.0	73.0	+1.7	−3.3
gpt-6-luna	gpt-6.1-terra	45.7	6.7	73.0	+5.3	+2.7	75.7	–	66.0	53.6	**+9.7**	**+22.1**
gpt-6-luna	Claude Haiku	45.7	6.7	66.8	+7.1	**+5.0**	71.8	69.7	61.4	53.6	**+10.3**	**+18.2**
gpt-6.1-terra	gpt-6-luna	29.7	4.3	53.6	+7.7	**+7.3**	60.9	60.6	66.7	73.0	−5.8	**−12.1**
gpt-6-luna	gpt-4.1	45.7	6.7	64.7	+7.7	**+6.3**	71.0	–	61.3	53.6	**+9.7**	**+17.4**
Claude Haiku	Claude Haiku	61.7	11.3	66.8	+9.8	**+5.7**	72.4	71.6	61.3	66.8	**+11.1**	**+5.7**
Claude Sonnet	Claude Haiku	65.0	9.7	66.8	+11.7	**+8.2**	75.0	73.8	60.4	75.7	**+14.6**	−0.7
gpt-6.1-terra	Qwen2.5-7B	31.0	4.0	50.3	+11.7	**+8.0**	58.3	55.0	62.3	73.0	−4.0	**−14.7**
gpt-6.1-terra	gpt-4o-mini	31.3	4.0	51.1	+12.8	**+9.6**	60.7	58.4	60.6	73.0	+0.1	**−12.3**
gpt-6-luna	gpt-6-luna	45.7	6.7	53.6	+13.8	**+11.1**	64.7	65.3	56.4	53.6	**+8.2**	**+11.1**
gpt-4o-mini	gpt-6-luna	67.0	30.3	53.6	+14.7	**+8.6**	62.1	61.9	56.1	51.1	+6.0	**+11.0**
gpt-4.1	gpt-6.1-terra	79.0	8.0	73.0	+15.7	**+7.3**	80.3	–	69.7	64.7	**+10.7**	**+15.7**
gpt-4o-mini	Qwen2.5-7B	67.0	30.3	50.3	+17.3	**+7.3**	57.7	54.7	53.3	51.1	+4.3	+6.6
gpt-4o-mini	gpt-4o-mini	67.0	30.3	51.1	+17.6	**+8.4**	59.6	57.1	52.4	51.1	+7.1	**+8.4**
gpt-6-luna	gpt-4o-mini	45.7	6.7	51.1	+19.1	**+13.7**	64.8	63.3	56.7	53.6	**+8.1**	**+11.2**
Claude Haiku	gpt-6-luna	61.7	11.3	53.6	+19.1	**+14.3**	67.9	68.7	63.0	66.8	+4.9	+1.1
gpt-6-luna	Qwen2.5-7B	48.0	6.7	50.3	+19.7	**+12.0**	62.3	59.7	58.0	53.6	+4.3	**+8.8**
gpt-4.1	Claude Haiku	79.0	8.0	66.8	+20.2	**+11.0**	77.8	76.0	63.7	64.7	**+14.1**	**+13.1**
gpt-4.1	gpt-4.1	79.0	8.0	64.7	+20.7	**+11.7**	76.3	–	63.3	64.7	**+13.0**	**+11.7**
Claude Haiku	gpt-4o-mini	61.7	11.3	51.1	+21.3	**+13.0**	64.1	62.7	57.9	66.8	+6.2	−2.7
Claude Sonnet	gpt-6-luna	65.0	9.7	53.6	+22.8	**+18.1**	71.7	72.9	67.8	75.7	+3.9	−4.0
Claude Sonnet	gpt-4o-mini	65.0	9.7	51.1	+28.1	**+15.9**	67.0	69.0	61.2	75.7	+5.8	**−8.7**
gpt-4.1	gpt-6-luna	79.0	8.0	53.6	+30.1	**+21.3**	74.9	76.8	61.2	64.7	**+13.7**	**+10.2**
gpt-4.1	gpt-4o-mini	79.0	8.0	51.1	+33.8	**+18.6**	69.7	67.6	57.9	64.7	**+11.8**	+5.0
gpt-4.1	Qwen2.5-7B	79.3	7.7	50.3	+36.7	**+16.0**	66.3	63.3	57.7	64.7	+8.7	+1.7

Table 35: Every detector–generator cell of H24 and H27 (ConflictBank, one-answer prompt, 300 conflict questions; %), sorted by the exploratory score. *Right*: the detector's hit rate, the share of questions on which it flags the true contradiction and names exactly its misinformation passage (computed on the detections that cell's generator saw; its direction precision, the same share among questions where it flags the true pair, pooled over its cells, is gpt-4o-mini 68.8%, gpt-4.1 81.8%, gpt-6-luna 82.2%, gpt-6.1-terra 82.6%, Claude Haiku 78.1% and Claude Sonnet 77.7%); *Wrong*: share on which it marks a correct passage as suspect; *Gen. acc.*: the generator's accuracy with no flags; *Score*: the mean over the 300 questions of [right direction and the generator errs] minus [a correct passage marked and the generator is right]; *Gain*: the LLM directional flag's change in points. *Verdict*: the detector's judgement passed as an answer key ("the answer X is false" for the answer its suspect passage supports; H30). *Hint*: the generator given the detector model's own plain answer, with no passage marked (H27); *det. alone*: the detector model's own plain accuracy. A detector's statistics can differ slightly between generators because concurrent runs made independent detector calls (up to 29 of 400 questions differ for one detector). Bold: significant under the paper's scheme (Gain: exploratory family; hint and det. alone columns: H27, Holm across cells).

Cell level (30 cells): accuracy with the flag, A	
$A \sim D + G$: coefficient of D (hit rate)	0.09 [0.03, 0.16]
coefficient of G	0.46 [0.36, 0.55]
$A \sim D$ + generator FE, permutation of D	$p = 0.29$
$A \sim P + R$ + gen. FE: precision P	1.03 [0.76, 1.20]; $p = 0.047$
recall R	0.16 [0.05, 0.26]; $p = 0.040$
Leave one detector out: R^2 for the gain	
G only	0.17
gap $D - G$ (registered)	-0.21
$P + R + G$	0.12
score (exploratory)	0.62
Question level, question/detector/generator FE (points)	
detector right on q	$+34.4$ [$+30.7$, $+38.3$]
detector marks a correct passage	-31.3 [-36.7, -26.5]

	H24 cells (observational)			H31 (randomised)	
Generator	right	wrong	wrong, abst.	right	wrong
Qwen2.5-7B	61	68	6	–	–
Claude Haiku	57	59	18	63	64
gpt-4.1	69	65	13	–	–
gpt-4o-mini	66	52	24	57	51
gpt-6-luna	75	57	16	72	77
gpt-6.1-terra	68	75	4	67	95

Table 36: H24 re-analysed (§15.4, §16.4; data already seen). D: hit rate (the detector flags the gold pair and names exactly its misinformation passage); P: direction precision (of those gold-pair flags, the share naming it); R: recall (the share of questions whose gold pair is flagged); G: the generator's no-flag accuracy. CIs: 2,000 question resamples shared across cells (300 with question fixed effects; the leave-one-detector-out range of the question-level coefficients is $+32.7$ to $+37.1$ and -32.8 to -26.8). p: exact permutation of the detectors' values across the six detectors; the design is unbalanced (the Claude detectors serve three generators), so detectors are only approximately exchangeable. Lower table, one definition of following: a right flag is followed when the flagged answer is correct (questions the generator gets wrong in every plain seed); a wrong flag is followed when the flagged answer contains the misinformation (questions it gets right in every plain seed); *abst.*: the flagged answer abstains instead. With no flag, a held-out plain seed changes outcome on at most 12% of such questions.

Policy	Setting	Neutral flag	Policy	Setting	Neutral flag
forced	CB forced, gpt-6.1-terra	−0.3	multi	RAMDocs, Claude Haiku	+1.2
forced	CB forced, gpt-4o-mini	−0.2	one	CB one-answer, gpt-4o-mini	−18.7
forced	CB forced, gpt-6-luna	+1.6	one	CB one-answer, gpt-6-luna	−9.4
forced	CB forced, Claude Haiku	+3.9	one	CB one-answer, gpt-4.1	−8.0
minimal	CB minimal, gpt-4o-mini	−18.1	one	CB majority, gpt-6.1-terra	−3.7
minimal	RAMDocs, gpt-4.1 (minimal)	−11.0	one	CB one-answer, Qwen2.5-7B	−3.3
minimal	RAMDocs, Qwen2.5-7B (minimal)	−5.8	one	CB one-answer, gpt-6.1-terra	−2.3
minimal	RAMDocs, gpt-4o-mini (minimal)	−5.3	one	RAGuard, gpt-6-luna	0.0
multi	CB multi-answer, gpt-4o-mini	−8.8	one	RAGuard, Claude Haiku	+0.3
multi	RAMDocs, gpt-4.1 (multi-answer)	−8.6	one	CB one-answer, Claude Haiku	+0.7
multi	CB multi-answer, Qwen2.5-7B	−8.3	one	CB majority, gpt-4o-mini	+1.1
multi	RAMDocs, gpt-4o-mini (multi-answer)	−6.7	one	RAGuard, gpt-6.1-terra	+2.4
multi	CB multi-answer, gpt-4.1	−6.3	one	CB one-answer, Claude Sonnet	+3.0
multi	RAMDocs, Qwen2.5-7B (multi-answer)	−5.8	one	RAGuard, gpt-4o-mini	+3.2
multi	RAMDocs, gpt-6-luna	−3.1	one	CB majority, Claude Haiku	+4.3
multi	RAMDocs, gpt-6.1-terra	−0.7	one	RAGuard, Claude Sonnet	+4.4
multi	RAMDocs, Claude Sonnet	0.0	one	CB majority, gpt-6-luna	+6.6
Multilevel model (random effects for generator, benchmark and setting; REML)					
pooled, all settings		−1.6 [−5.5, +2.2]; prediction interval [−13.6, +10.3]			
pooled, multi-answer prompt (10 settings)		−3.8 [−6.5, −1.0]			
pooled, minimal prompt (4 settings)		−12.1 [−23.6, −0.7]			
pooled, one answer prompt (16 settings)		−0.5 [−5.5, +4.5]			
pooled, forced prompt (4 settings)		+1.5 [−1.3, +4.3]			
Meta-regression (same model)					
intercept (OpenAI, one answer, generated)		−2.8 [−8.1, +2.6]			
several answers allowed		−3.5 [−8.2, +1.2]			
forced (no unknown)		+6.4 [+1.6, +11.2]			
Anthropic		+5.2 [+1.7, +8.7]			
Qwen		+1.0 [−4.0, +6.0]			
natural misinformation (RAGuard)		+2.7 [−7.0, +12.3]			

Table 37: The 34 settings in which the oracle neutral flag was run (conflict questions, change in points), grouped by answer policy. The pooled estimate was registered as exploratory (§13.4) and the multilevel model, the policy split and the moderators in §16.4; they use data already seen. The prediction interval describes this design space (our generators, benchmarks and prompts), not deployments in general. *Natural misinformation* is RAGuard only; RAMDocs' misinformation is synthetic. The moderator *several answers allowed* codes the multi-answer and minimal prompts together, against one-answer prompts; the policy rows above separate them. Settings share questions, which the model does not account for, so its intervals are too narrow.

	Neutral flag, utility change at misinformation cost c				Dir. flag	Neutral flag: plain answer → flagged answer (null)			
Setting	$c=0$	$c=1$	$c=2$	$c=4$	$c=2$	mis.→abst.	mis.→corr.	corr.→abst.	corr.→mis.
RAMDocs, gpt-4o-mini (multi-answer)	**−6.7**	**−20.9**	**−35.0**	**−63.3**	**+79.1**	1 (0)	2 (2)	0 (0)	29 (3)
RAMDocs, gpt-6-luna	**−3.1**	−7.7	−12.2	−21.3	**+160.9**	0 (0)	1 (3)	0 (0)	22 (16)
RAMDocs, Claude Haiku	+1.2	+9.1	**+17.0**	**+32.9**	**+125.2**	0 (0)	9 (4)	0 (0)	5 (9)
CB multi-answer, gpt-4o-mini	**−8.8**	**−19.2**	**−29.7**	**−50.6**	**+91.0**	0 (0)	4 (3)	1 (0)	55 (9)
CB one-answer, gpt-4o-mini	**−18.7**	−5.3	+8.0	**+34.7**	**+83.7**	45 (2)	7 (3)	37 (1)	5 (2)
CB one-answer, gpt-6-luna	**−9.4**	−2.6	+4.3	+18.1	**+85.1**	27 (10)	12 (13)	20 (6)	5 (6)
CB one-answer, gpt-6.1-terra	−2.3	0.0	+2.3	+7.0	**+51.7**	2 (–)	26 (–)	4 (–)	5 (–)
CB one-answer, Claude Haiku	+0.7	+0.6	+0.4	+0.2	**+66.9**	0 (0)	25 (16)	0 (0)	8 (5)
CB one-answer, Claude Sonnet	+3.0	+6.7	+10.3	+17.7	**+45.7**	2 (–)	34 (–)	0 (–)	3 (–)
CB forced, gpt-4o-mini	−0.2	−0.6	−0.9	−1.6	**+74.0**	1 (0)	19 (4)	0 (0)	11 (2)
CB majority, gpt-4o-mini	+1.1	**+24.6**	**+48.0**	**+94.9**	**+118.2**	26 (1)	11 (2)	22 (1)	8 (5)
RAGuard, gpt-4o-mini	+3.2	+6.6	+10.0	+16.8	**+34.6**	12 (3)	7 (3)	8 (6)	4 (2)
RAGuard, gpt-6-luna	0.0	−1.1	−2.3	−4.5	**+29.3**	10 (9)	6 (9)	8 (9)	7 (8)
RAGuard, Claude Haiku	+0.3	−1.8	−4.0	−8.3	**+26.5**	7 (6)	4 (4)	11 (9)	6 (3)

Table 38: Abstention-aware view of the oracle flags (conflict questions; §15.4 and §16.4, re-analyses of data already seen). Utility per answer: +1 correct, $-c$ if it contains the misinformation, 0 for abstaining or any other wrong answer; $c=0$ is strict accuracy. Bold: significant in one Holm family per c over all 34 settings of Table 37 (a selection is shown). Right: of the question–seed pairs whose plain answer contained the misinformation (or was correct), the share whose flagged answer abstained, was correct or contained the misinformation; in brackets the same share between two plain runs with different seeds, the churn with no flag.